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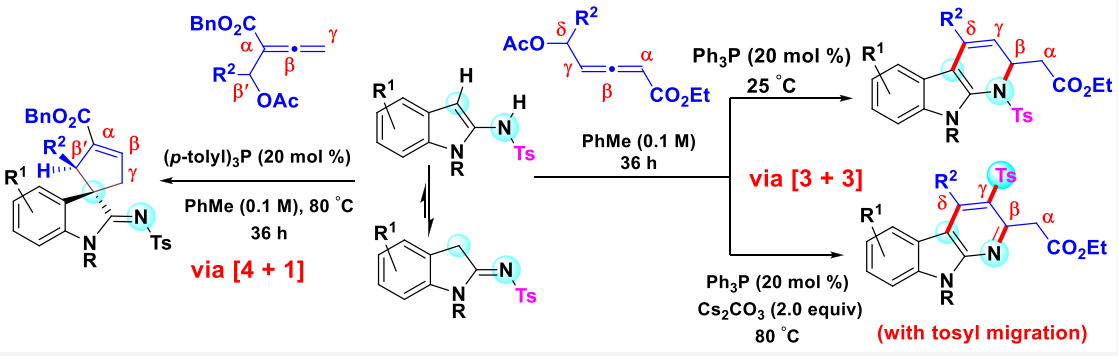
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Divergent Reactivity of δ- and β′-Acetoxy Allenoates with 2-Sulfonamidoindoles via Phosphine Catalysis: Entry to Dihydro-α-carboline, α-Carboline, and Spiro-cyclopentene Motifs

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ABSTRACT: The reactivity of 2-sulfonamidoindoles with acetoxy allenoates under phosphine catalysis depends on the disposition of the acetoxy (OAc) group on the allenoate. In the temperature-controlled [3 + 3] annulations, δ-acetoxy allenoates afforded dihydrocarboline and carboline scaffolds with carbon−nitrogen nucleophilic 2-sulfonamidoindoles, in which allenoate serves as a β-, γ-, and δ-carbon donor. At room temperature (25 °C), dihydro-α-carboline motifs were obtained exclusively through Michael addition, 1,4-proton shift, isomerization, 1,2-proton transfer, phosphine elimination, and aza-Michael addition. The higher temperature (80 °C) cascade protocol using Ph_3P−Cs_2CO_3 combination involves addition−elimination, aza-Claisen rearrangement, *tosyl migration*, and aromatization as key steps to give α-carbolines containing *tosyl functionality at the γ-carbon*. In contrast, with β′-acetoxy allenoate, 2-sulfonamidoindole acts only as a carbo-nucleophile in $(p\text{-tolyl})_3P$-directed [4 + 1] spiro-annulation, leading to five-membered *spiro-carbocyclic motifs* essentially as single diastereomers (dr >20:1) via chemoselective carbo-annulation.

INTRODUCTION

Over the past few decades, transformations involving allenes have evolved as powerful tools in organic synthesis.[1,2] Since the discovery of Lu's [3 + 2] cycloaddition of allenoates using phosphine,[3] numerous Lewis base-catalyzed/mediated annulations of allenoates have been reported.[4,5] Later, acetoxy allenoates were widely utilized as useful synthons in a variety of annulations by Tong as well as other groups.[6,7] Generally, the nucleophilic attack of Lewis base at sp-hybridized carbon generates a zwitterionic intermediate, which is then trapped *in situ* with other partners. These annulations are demonstrated as reliable approaches in organic synthesis leading to a wide array of carbo- and heterocycles (Scheme 1). Tong and co-workers reported a [3 + 3] annulation with δ-acetoxy allenoate functioning as 3C-synthon (α-, β-, and γ-carbons participated) using phosphine (Scheme 1b).[8a] Subsequently, Tong's[8b,e] and Shi's[8c] groups successfully explored [3 + 2] annulations (Scheme 1c) of δ-acetoxy allenoates with 1C, 2/3N binucleophiles using phosphine; these reactions were facilitated by the addition of nucleophile at the δ-carbon of diene-phosphonium intermediate. In 2017, the Tong's group reported the reaction between δ-acetoxy allenoate and 2-naphthol in which allenoate acted as a 2C (γ and δ) synthon, but in the case of 1-naphthol, the allenoate acted as both a 2C (γ and δ) and 3C- (α, β, and γ) synthon and led to five- as well as six-membered oxacycles (Scheme 1d).[8d] Very recently, Zhou's group[8f] reported a domino [3 + 2] and [3 + 2] annulation protocol involving δ-acetoxy allenoate as a 2C-synthon (γ, δ or β, γ) to synthesize a fused six-membered heterocyclic core with an exocyclic double bond (Scheme 1e). Apart from these, previously, Tong's group reported the first example of [4 + 1] annulation using a β′-acetoxy allenoate leading to a five-membered carbocyclic motif (Scheme 1f)

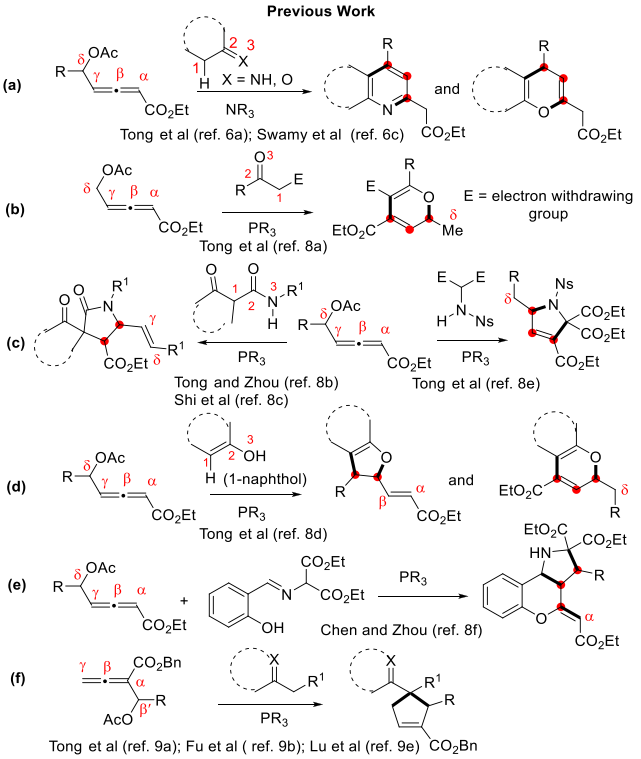
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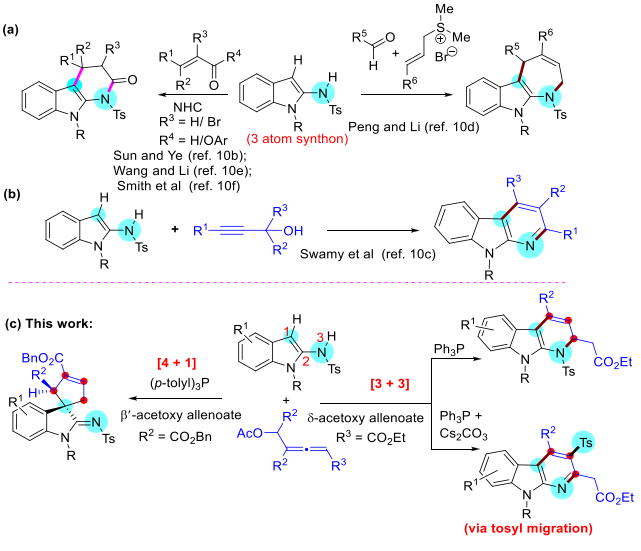
Scheme 1. Lewis Base Catalyzed/Mediated Annulation of Acetoxy Allenoates



under phosphine catalysis.[9a] Subsequently, others also employed β'-acetoxy allenoate in numerous [4 + 1] annulations,[9b−e] but spiro-carbo annulations are scarce.[9f] However, to the best of our knowledge, [3 + 3] annulation involving β-, γ-, and δ-carbons of δ-acetoxy allenoates as well as *functional group migration* using phosphine has not been reported until now.

Previously, 2-sulfonamidoindoles have been used as versatile building blocks for the construction of six-/seven-membered aza-heterocyclic cores (Scheme 2a,b),[10] but not for the generation of the spirocyclic motifs or 1,2-dihydrocarbolines. Spirocyclic quaternary stereocenters have gained a great deal of attention in organic synthesis[11] and often appear in bioactive compounds,[12] while α-carbolines are encountered in a variety of bioactive molecules.[13] As a part of our continued interest in the reactions of allenes,[1g,6c,d,14] herein we disclose chemoselective [3 + 3]-hetero- and [4 + 1] spiro-carbo-annulations of 2-sulfonamidoindoles with acetoxy allenoates that highlight (i) Ph_3P-catalyzed temperature-dependent [3 + 3] heteroannulations of δ-acetoxy allenoates affording a novel class of dihydro-α-carboline and α-carboline motifs, the latter via tosyl migration to the γ-carbon, in which sulfonomidoindole acts as a C/N-donor and (ii) $(p\text{-tolyl})_3P$-catalyzed [4 + 1] spiro-carbo-annulation of β'-acetoxy allenoates leading to spirocyclopentene motifs with excellent stereoselectivity (dr up to 20:1).

Scheme 2. Synthesis of Aza and Carbocyclic Cores Using 2-Sulfonamidoindoles



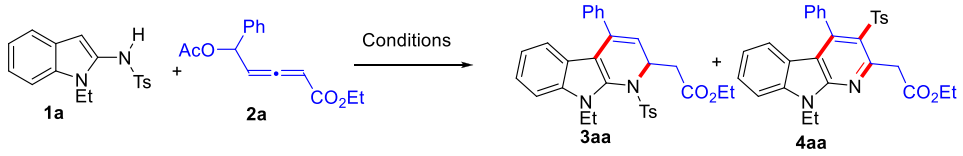
RESULTS AND DISCUSSION

To optimize the reaction conditions, we explored the reaction of 2-sulfonamidoindole **1a** with δ-acetoxy allenoate **2a**. The results are collated in Table 1. Initially, we tested the reaction between **1a** (0.20 mmol) and **2a** (0.24 mmol) in the presence of Ph_3P (20 mol %) in toluene at rt (25 °C), but we obtained a product **3aa** in 45% yield after 24 h (entry 1). The yield (67%) of the **3aa** could be enhanced by using degassed toluene (entry 2). We wish to point out that we missed this point earlier and had used 1 equiv of Ph_3P to get the maximum yield. Satisfyingly, we isolated the α-carboline **3aa** in 83% yield (entry 3) after 36 h. Several other phosphines were screened, but all of them were less effective (entries 4−7). Both electron-poor phosphines [$(p\text{-}FC_6H_4)_3P$ and $(p\text{-}ClC_6H_4)_3P$] and electron-rich phosphines [$MePPh_2$ and $(n\text{-}Bu)_3P$] gave lower yields compared to Ph_3P. Hence, we believe that optimum electron density and steric factors around phosphorus are crucial for this reaction. Further screening of reaction conditions using different solvents (DCE, DMF, MeOH, 1,4-dioxane, and H_2O; entries 8−12) revealed that toluene was the best solvent for this annulation, providing **3aa** in 83% yield (entry 3). To further increase the yield, we performed a reaction at a higher temperature (80 °C) but found a completely different product **4aa** (<10%) along with **3aa** (61%); we identified product **4aa** as the tosyl migrated and aromatized carboline (entry 13) using analytical data (vide infra). Inspired by this unexpected result, we performed several reactions to obtain **4aa** as the sole product by minimizing the yield of **3aa**. First, we tested a reaction by using Cs_2CO_3 as an additive; fortunately, **4aa** was obtained in 75% yield (entry 14). Other additives (K_2CO_3 and ^{t}BuOK) were less effective (entries 15 and 16) compared to Cs_2CO_3. At room temperature, the $Ph_3P-Cs_2CO_3$ combination did not give a clear-cut reaction leading to either **3aa** or **4aa** (entry 17). Also, low catalytic loading (10 mol %) of phosphine did not lead to reactions efficiently (entries 18 and 19). Use of 50 or 100 mol % of phosphine gave essentially the same yield as that from 20 mol % (entries 20−23).

After establishing the best reaction conditions, we investigated the scope and limitations of the [3 + 3] annulation by using several 2-sulfonamidoindoles as C/N-donors and a range of δ-acetoxy allenoates as 3C-synthons (β, γ, δ-carbon donor). As shown in Scheme 3, regardless of the nature of substituents (OMe, Cl, or CF_3) at the third/fourth position on the phenyl ring of allenoate (**2a−2e**), good reactivity with sulfonamidoindole **1a** was achieved and the reaction afforded

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Table 1. Optimization of Reaction Conditions for 3aa and 4aam[a,b]



entry	base	additive	solvent	temp (°C)	time (h)	yield[b] **3aa**/**4aa** (%)
1	Ph_3P		PhMe	25	24	45:00
2	Ph_3P		PhMe	25	24	67:00
3	**Ph_3P**		**PhMe**	25	**36**	**83:00**
4	$(p\text{-}ClC_6H_4)_3P$		PhMe	25	36	61:00
5	Bu_3P		PhMe	25	36	57:00
6	$(p\text{-}FC_6H_4)_3P$		PhMe	25	36	58:00
7	Ph_2PMe		PhMe	25	36	65:00
8	Ph_3P		DCE	25	36	26:00
9	Ph_3P		DMF	25	36	29:00
10	Ph_3P		MeOH	25	36	
11	Ph_3P		Dioxane	25	36	31:00
12	Ph_3P		H_2O	25	36	
13	Ph_3P		PhMe	80	36	61:<10
14	**Ph_3P**	**Cs_2CO_3**	**PhMe**	**80**	**36**	**≤5:75**
15	Ph_3P	K_2CO_3	PhMe	80	36	≤5:61
16	Ph_3P	^{t}BuOK	PhMe	80	36	≤5:64
17	Ph_3P	Cs_2CO_3	PhMe	80	36	Complex
18	Ph_3P	Cs_2CO_3	PhMe	80	36	≤5:53
19	Ph_3P		PhMe	25	36	61:00
20	Ph_3P		PhMe	25	36	83:00
21	Ph_3P	Cs_2CO_3	PhMe	80	36	≤5:75
22	Ph_3P		PhMe	25	36	83:00
23	Ph_3P	Cs_2CO_3	PhMe	80	36	≤5:75

[a]Reaction conditions: **1a** (0.20 mmol), **2a** (0.24 mmol), phosphine (0.04 mmol for entries 1−18; 0.02 mmol for entries 18 and 19; 0.10 mmol for entries 20 and 21; 0.20 mmol for entries 22 and 23), and additive (0.40 mmol for entries 14−18, 21 and 23) in solvent (2.0 mL). [b]Isolated yield.

dihydro-α-carbolines **3aa**−**3ae** in 76−85% yields. Similarly, disubstituted allenoate **2f** also reacted efficiently to afford **3af** in 75% yield. The δ-polyaryl-substituted allenoate **2g** was completely consumed to deliver **3ag** in decent yield (72%). δ-Thienyl- and indolyl-substituted allenoates **2h** and **2i** gave the desired products **3ah** and **3ai** in yields of 73−75%. Moreover, this methodology was also achieved with aliphatic allenoate **2j** to deliver the [3 + 3]-annulated product **3aj** in 71% yield. Different substituents on the nitrogen atom in the nucleophile (**1b**−**d**) were well-tolerated affording high yields (73−84%) of **3ba**−**3da**. Finally, this methodology was also amenable with 5- or 6-substituted sulfonamidoindoles (**1e** and **1f**), providing dihydrocarbolines **3ea** and **3fa** in good yields (74% and 71%).

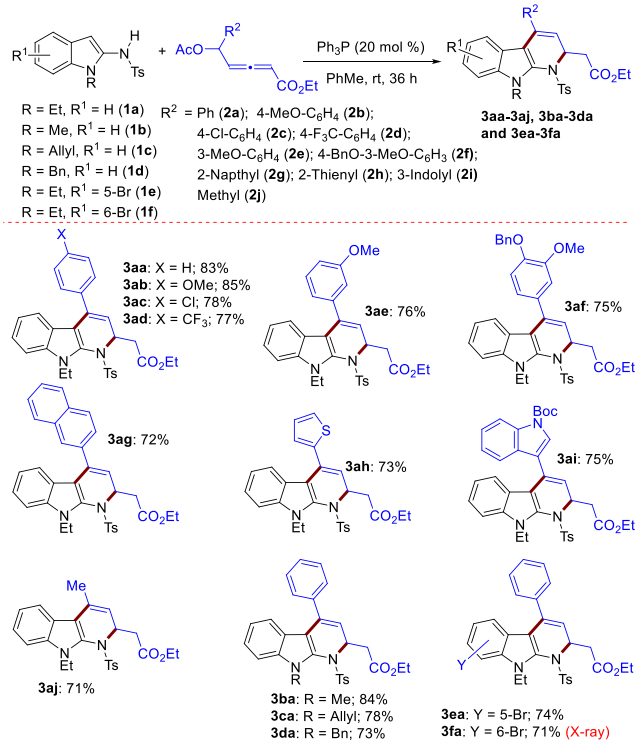
Next, we explored the substrate scope involving δ-acetoxy allenoates and sulfonamidoindoles in annulation and tosyl migration cascade processes leading to products of type **4**. As indicated in Scheme 4, allenoates **2a**−**2d** afforded products **4aa**−**4ad** in 70−78% yields. To our delight, allenoate derived from vanillin **2f** and the heterocyclic allenoate **2h** also provided carbolines **4af** and **4ah** in good yields (68% and 64%, respectively). δ-Methyl-substituted allenoate **2j** reacted nicely in the cascade process and afforded **4aj** in good yield (68%). Allenoate with naphthyl group at the fifth position **2k** also showed decent reactivity with **1a** to lead to the corresponding product **4ak** in 72% yield. Finally, *N*-substituted and 5-substitiuted sulfonamidoindoles (**1b** and **1e**) also afforded **4ba** and **4ea** in decent yields (76% and 65%) with **2a**. It is known that Cs_2CO_3 has fair solubility in many organic solvents and quite low solubility in toluene, but the dramatic effect of tosyl group transfer as observed here is rather unexpected.[15]

Taking a cue from the reactivity of δ-acetoxy allenoates as described above, we expected the formation of seven-membered fused heterocycle using β'-acetoxy allenoates. Thus, as depicted in Table 2, we performed a reaction between 2-sulfonamidoindole **1a** (0.20 mmol) and β'-acetoxyallenoate **5a** (0.24 mmol) using the conditions shown for entry 3, Table 1. Surprisingly, we isolated product **6aa** in 47% yield and identified it as the [4 + 1] product via spiro-annulation but not [4 + 3]-annulated product **7aa** (Table 2, entry 1). This result motivated us to investigate the [4 + 1] annulation reaction in a systematic way for improving the yield of **6aa**. Increasing the reaction temperature from 25 to 80 °C resulted **6aa** in 69% yield (entry 2). Investigation of several phosphines (entries 3−7) indicated that $(p\text{-tolyl})_3P$ was the best choice as a catalyst (entry 7). We then screened other parameters including solvent and additive. Solvent study showed that toluene to be the best (entry 7); protic polar and aprotic solvents (entries 8−11) gave inferior results. However, addition of Cs_2CO_3 did not show any dramatic effect in the reactivity (entry 12). Thus, conditions under entry 7 were found to be the best for this [4 + 1]-spiro-annulation.

As depicted in Scheme 5, various β'-acetoxy allenoates **5** reacted efficiently with **1a** under the standard conditions to afford the corresponding spirocyclic cyclopentene carboxylate motifs with excellent diastereoselectivity (dr up to >20:1). The substituents on the phenyl ring (**5a**−**d**) had no significant effect on this methodology, and spirocyclics **6aa**−**6ad** were

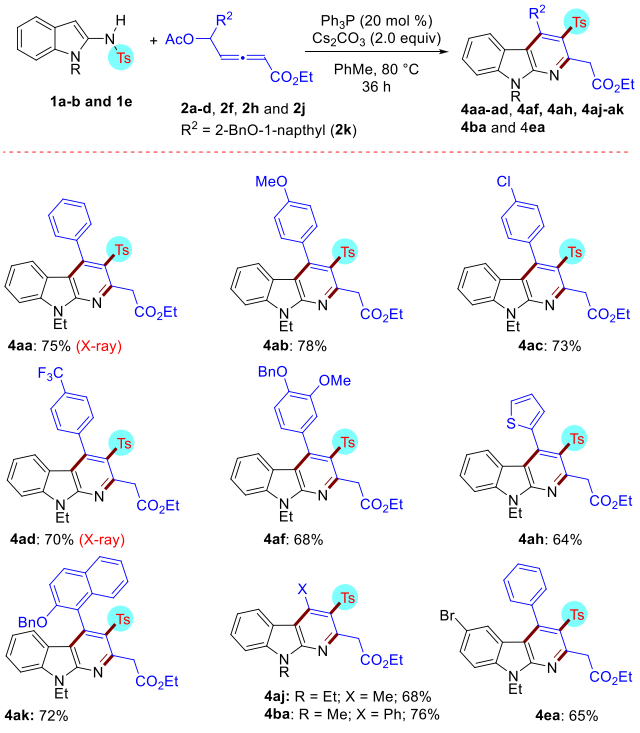
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Scheme 3. Scope of [3 + 3] Hetero-annulation[a,b]



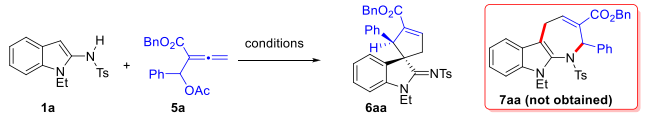
[a]Reaction conditions: **1a–f** (0.20 mmol), **2a–j** (0.24 mmol), and Ph_3P (0.04 mmol) in toluene (0.1 M) at room temperature (25 °C). [b]Isolated yield.

Scheme 4. Scope of [3 + 3]-Annulation Accompanied by Tosyl Migration[a,b]



[a]Reaction conditions: **1a,b** or **1e** (0.20 mmol), **2a–d**, **2f**, **2h**, **2j**, or **2k** (0.24 mmol), Ph_3P (0.04 mmol), and Cs_2CO_3(0.40 mmol) in toluene (0.1 M) at 80 °C. [b]Isolated yield.

Table 2. Optimization of Reaction Conditions for 6aa[a–c]



entry	base	solvent	temp (°C)	time (h)	yield[b] of **6aa** (%)	dr of **6aa**[c]
1	Ph_3P	PhMe	25	36	47	20:1
2	Ph_3P	PhMe	80	36	69	20:1
3	Ph_2PMe	PhMe	80	36	54	20:1
4	$(p\text{-}ClC_6H_4)_3P$	PhMe	80	36	49	20:1
5	Bu_3P	PhMe	80	36	52	20:1
6	$(p\text{-}FC_6H_4)_3P$	PhMe	80	36	45	20:1
7	**$(p$-tolyl$)_3$P**	**PhMe**	**80**	36	**79**	20:1
8	$(p$-tolyl$)_3$P	MeCN	80	36	26	20:1
9	$(p$-tolyl$)_3$P	DMF	80	36	29	20:1
10	$(p$-tolyl$)_3$P	MeOH	80	36	trace	
11	$(p$-tolyl$)_3$P	H_2O	80	36	trace	
12	$(p$-tolyl$)_3$P	PhMe	80	36	79	20:1

[a]Reaction conditions: **1a** (0.20 mmol), **2a** (0.24 mmol), phosphine (0.04 mmol), and Cs_2CO_3 (0.40 mmol for entry 12) in solvent (2.0 mL). [b]Isolated yield. [c]Diastereomeric ratio.

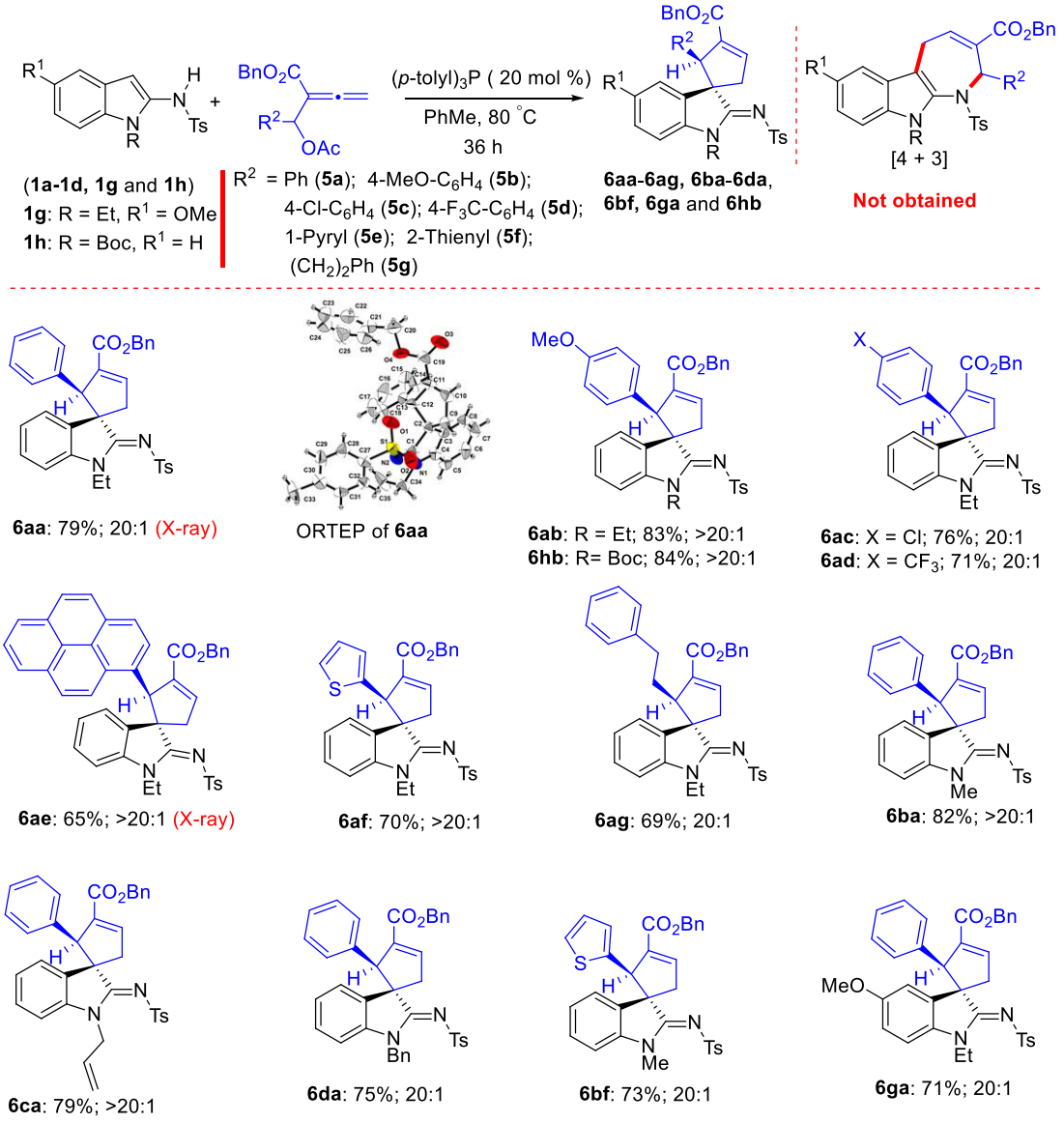
isolated in 71–83% yields. Even the pyrenyl allenoate **5e** and heterocyclic allenoate **5f** reacted well with **1a** and delivered **6ae** and **6af** in yields of 65% and 70%, respectively. Aliphatic allenoate **5g** also reacted nicely in the spiro-annulation to lead **6ag** in 69% yield. Substituted indolines (**1b–d**, **1g**, and **1h**) were also suitable for this reaction and gave spirocyclic motifs **6ba–6da**, **6bf**, **6ga**, and **6hb** in good to high yields (75–82%, 73%, 71%, and 84%). The relative configuration of the substituents in these products was ascertained by the X-ray crystal structures of **6aa** and **6ae**. The allenol precursors corresponding to the acetates **2a** and **5a** did not undergo any perceptible reaction even after 48 h, suggesting that acetate leaving group is essential. In addition, the use of (*S*)-BINAP as the phosphine did not give product **6aa**; although derivative **3aa** could be obtained, chiral induction did not occur.

In the reaction of *N*-Boc-protected 2-sulfonamidoindole (**1h**) with acetoxy allenoate **2a** under optimized conditions (Table 1, entry 3), surprisingly, we observed that the noncyclized dienolate **7ha** as the sole product via α-umpolung addition (Scheme 6).

It can be noted that the aromatized tosyl-migrated product **4aa** has two hydrogen atoms missing when compared to **3aa**. To investigate the mechanism of tosyl migration cum [3 + 3] annulation, we performed a few experiments. Heating compound **3aa** individually with (i) Cs_2CO_3, (ii) Ph_3P, or (iii) Ph_3P+Cs_2CO_3 at 80 or 110 °C even for 36 h did not yield compound **4aa** (Scheme 7a). Thus, it is likely that [3 + 3] annulation *with and without* tosyl migration occurs by different pathways. Also, in the absence of Ph_3P, we did not observe any product at rt or at 80 °C (Scheme 7b). The yield of **4aa** dramatically dropped when we used tosyl chloride as tosyl cation source (Scheme 7c). Based on these experiments, we infer that (i) the presence of Ph_3P is essential, (ii) temperature and gradual addition of allenoate play a key role, and (iii) tosyl anion (not cation) is most likely involved in the migration leading to **4aa**.

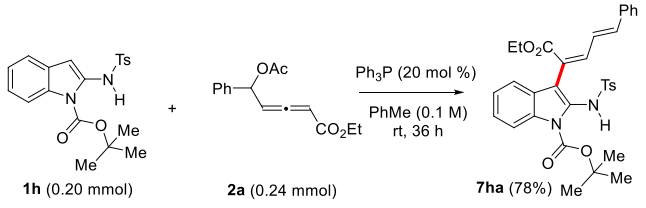
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Scheme 5. Scope of [4 + 1] Spiro-Carbo-Annulation[a−c]



[a]Reaction conditions: **1a−d**, **1g−h** (0.20 mmol), **5a−g** (0.24 mmol), and (*p*-tolyl)$_3$P (0.04 mmol) in toluene (0.1 M) at 80 °C. [b]Isolated yield. [c]Diastereomeric ratio.

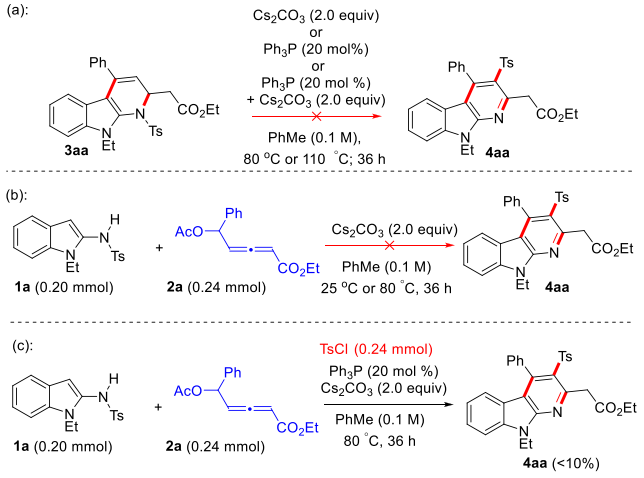
Scheme 6. Isolation of Dienolate 7ha from the Reaction of *N*-Boc-sulfonamidoindole (1h) with *δ*-Acetoxy Allenoate (2a)



Plausible pathways for the [3 + 3] and [4 + 1] annulations are shown in Scheme 8. These divergent annulations may be initiated through the phosphonium intermediate **A1** or **C1** via the addition−elimination process between phosphine and **2a** or **5a**. In the [3 + 3] annulation, the addition of **1a′** (generated from **1a**) to the phosphonium ion **A1** at the *δ*-carbon gives zwitterionic intermediate **A2**. Immediately **A2** undergoes 1,4-proton transfer and affords **A3**. Subsequently, **A3** upon double bond isomerization, 1,2-proton transfer, and phosphine elimination leads to dienoate **A6** via **A4** and **A5**. Finally, **A6** after intramolecular aza-Michael addition[6c] yields **3aa**. For the formation of tosyl migration product **4aa**, one possible pathway is shown in Scheme 8b. In this **A1** undergoes a second addition−elimination process with **1a′** to give the allenic intermediate **B3**. Intermediate **B3** is involved in the aza-Claisen rearrangement[15,16a] to give zwitterionic intermediate **B4**. Finally, tosyl group elimination/readdition, and aromatization involving species **B4−B7** are expected to be involved in the formation of the unexpected *α*-carboline with a tosyl functionality at the *γ*-carbon **4aa**. It is possible that the nitrogen atom of intermediate **B3** is electron deficient due to the double bond and sulfonyl functionality.[15] Hence, one of the double bonds of the allene moiety attacks this nitrogen to deliver zwitterionic intermediate **B4**. Furthermore, at rt, proton

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Scheme 7. Control Experiments



shift is faster than addition−elimination, but at 80 °C addition−elimination is faster than proton shift. Therefore, the two [3 + 3] leading to **3aa** and **4aa** annulations occur in distinct ways. In the (p-tolyl)$_3$P-mediated [4 + 1] stereoselective spiro-annulation (Scheme 8c), nucleophile **1a′** selectively attacks at the γ-carbon of **C1** to offer the ylide intermediate **C2**. Then **C2** is involved in 1,3-proton transfer to produce **C3**. Finally, **C3** undergoes addition−elimination (S_N2'-attack) leading to **6aa**. It should be noted that in the [4 + 1] spiro-carbo-annulation the key step is ylide formation and the formation of the C−C bond is more favorable than the C−N bond in the final step. Although the possibility of the formation of a 7-membered ring exists (cf. structure **C3**), we have not observed it in the present study. Another possible pathway (based on a reviewer's comment) for [4 + 1] spiro-carbo-annulation is given in the Supporting Information (Scheme S1).

The robustness of our divergent annulations is exemplified by 1.0 mmol scale reactions of **1a** with **2a** or **5a** under the optimal conditions to afford **3aa**, **4aa**, and **6aa** in high yields (Scheme 9).

CONCLUSIONS

We have developed phosphine-catalyzed divergent annulations that utilize β'/δ-acetoxy allenoates (as carbo-electrophiles) and 2-sulfonamidoindoles. In the temperature-dependent phosphine-catalyzed [3 + 3] annulations, δ-acetoxy allenoates act as β-, γ-, and δ-carbon donors with 2-sulfonamidoindoles. At room temperature, we obtained medicinally significant 1,2-dihydrocarbolines in good to high yields. In sharp contrast, tosyl migration and aromatization/cascade process occurred at higher temperature (80 °C) to deliver α-carboline motifs with tosyl functionality at the γ-carbon. In the stereoselective [4 + 1] spiro-carbo-annulation, 2-sulfonamidoindole acts as a 1C synthon with β'-acetoxy allenoates to provide a novel class of spirocyclopentene carboxylates with excellent diastereomeric ratio (dr up to >20:1).

EXPERIMENTAL SECTION

General Methods. All of the reactions were carried out either under an inert atmosphere or in air. All required chemicals were procured from Aldrich or local manufacturers and used as such without further purification, unless noted. Melting points have been recorded on MR-visual melting range apparatus and are uncorrected. ^{1}H, ^{13}C{^{1}H}, and ^{19}F NMR spectra were recorded using 5 mm tubes on a 400 and 500 MHz NMR spectrometer [field strengths: for ^{13}C{^{1}H} NMR 400/100 or 500/125 MHz and for ^{19}F NMR 400/376 or 500/470 MHz respectively] in $CDCl_3$ solution (unless specified otherwise) with shifts referenced to TMS (^{1}H, ^{13}C: = 0) or $CFCl_3$ (^{19}F: = 0). Data for ^{1}H, ^{13}C{^{1}H}, and ^{19}F NMR are reported in δ (ppm). All J values are reported in hertz. Infrared spectra were recorded using an ATR technique on a FT-IR spectrophotometer. Mass spectra were recorded using HRMS (ESI-Thermo Scientific EXACTIVE ORBITRAP/TOF analyzer) equipment. Crystallographic data were collected on a X-ray diffractometer system using Mo Kα radiation (λ = 0.71073 Å) radiation. Structures were solved and refined using standard methods. Thin-layer chromatography was performed on silica/alumina plates and components were visualized under iodine/UV light at 254 nm. Column chromatography was performed on silica gel (100−200 mesh), for column elution process hexane-EtOAc mixture was used as the eluent unless otherwise stated.

The starting materials **1a−g** were prepared by using the procedure of Kumara Swamy et al.,[10c] and **1h** was prepared by the method of Chang's group.[17] Allenoate precursors **2a−k**[6c] and **5a−g**[18] were synthesized using literature procedures. Of these, **1a−e** and **2a−j** are known; precursors **1f**, **1h**, and **5a−g** are new. The solvent of crystallization for all products was ethyl acetate−hexane (ca. 4:1 v/v).

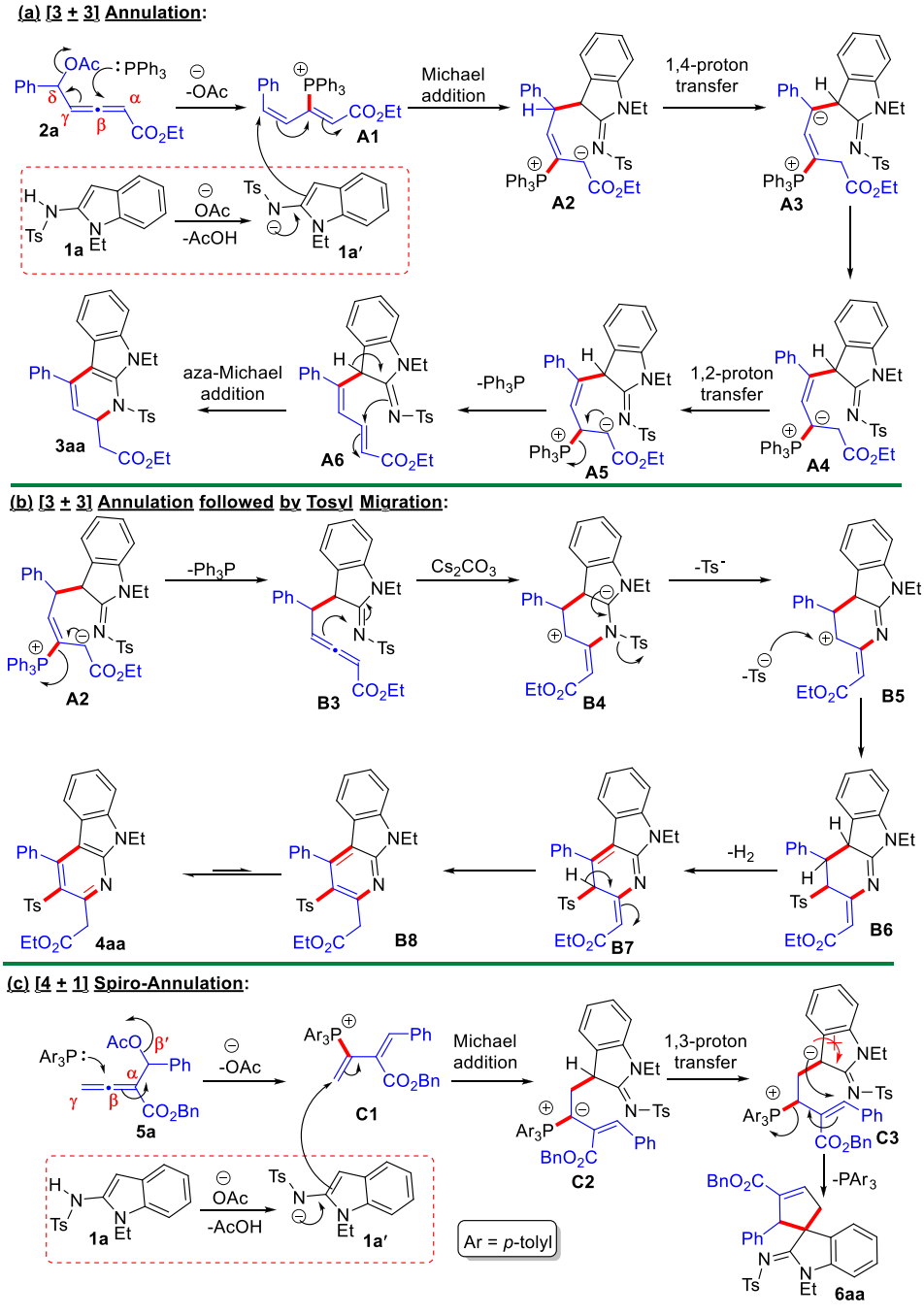
*i. Synthesis of N-(6-Bromo-1-ethylindolin-2-ylidene)-4-methylbenzenesulfonamide (**1f**).* An oven-dried 25 mL round-bottomed flask was charged with 6-bromo-1-ethyl-1*H*-indole (1.11 g, 5.0 mmol) and *p*-toluenesulfonyl azide (1.38 g, 7.0 mmol) in 10.0 mL of dry 1,4-dioxane under nitrogen atmosphere. The mixture was stirred with heating (oil bath temperature at 75−80 °C) for 18−24 h and cooled. Addition of ethanol (40 mL) caused most of the 2-sulfonamidoindoline **4** to crystallize/precipitate from the solution. The solvent was removed from the mother liquor and the residue purified by column chromatography on silica gel using ethyl acetate/hexane (1:4) as eluent to obtain 2-sulfonamidoindole **1f** as a brown solid. Yield: 1.17 g (60%), mp 137−139 °C. ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.89−7.88 (m, 2H), 7.31 (d, J = 7.5 Hz, 2H), 7.25 (dd, J = 8.0, 2.0 Hz, 1H), 7.20−7.19 (m, 1H), 7.11−7.10 (m, 1H), 4.17 (s, 2H), 3.89 (q, J = 7.5 Hz, 2H), 2.43 (s, 3H), 1.27 (t, J = 7.5 Hz, 3H). ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 169.1, 144.5, 142.9, 139.2, 129.4, 126.7, 126.3, 125.7, 125.5, 121.6, 112.6, 37.1, 35.9, 21.5, 11.9. IR (neat): ν_{max} 3056, 2985, 2945, 2907, 2306, 1735, 1566, 1445, 1374, 1266, 1246, 1156, 1102, 1046, 943, 897, 848, 743, 705, 665 cm^{-1}. HRMS (ESI-TOF): m/z [M + H]$^+$ calcd for $C_{17}H_{18}BrN_2O_2S$ 393.0267, found: 393.0261.

*ii. Synthesis of tert-Butyl 2-((4-Methylphenyl)sulfonamido)-1H-indole-1-carboxylate (**1h**).* An oven-dried 25 mL round-bottomed flask was charged with *tert*-butyl 1*H*-indole-1-carboxylate (500 mg, 2.30 mmol), 4-methylbenzenesulfonyl azide(500 mg, 2.54 mmol), $AgNTf_2$(89.2 mg, 0.23 mmol), and AgOAc (76.7 mg, 0.26 mmol) in 5.0 mL of DCE. The mixture was stirred with heating (oil bath temperature at 40 °C) for 12 h and cooled. The reaction mixture was cooled to room temperature and filtered through a plug of Celite and then washed with EtOAc. The solvent was removed from the mother liquor and the residue purified by column chromatography on silica gel using ethyl acetate/hexane (1:4) as the eluent to obtain 2-sulfonamidoindole **1h** as a white solid. Yield: 100.0 mg (13%), mp 145−147 °C. ^{1}H NMR (500 MHz, $CDCl_3$): δ 9.65 (s, 1H), 7.77−7.76 (m, 3H), 7.44 (d, J = 7.0 Hz, 1H), 7.24−7.16 (m, 4H), 6.56 (s, 1H), 2.38 (s, 3H), 1.65 (s, 9H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ 151.8, 144.2, 136.0, 133.7, 132.4, 129.7, 128.5, 127.4, 123.6, 123.2, 120.2, 115.2, 96.8, 86.2, 28.1, 21.6. IR (neat): ν_{max}3244, 2979, 2931, 1705, 1598, 1453, 1371, 1330, 1210, 1157, 1089, 886, 665 cm^{-1}. HRMS (ESI-TOF): m/z [M + H]$^+$ calcd for $C_{20}H_{23}N_2O_4S$ 387.1373, found 387.1376.

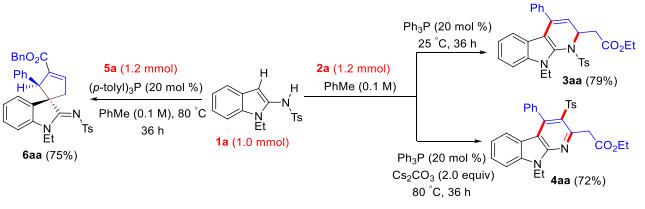
*iii. General Procedure for the Synthesis of Compounds **5a−g**.* Following a literature procedure,[18] to a solution of benzyl 2-(hydroxy-4-aryl/heteroaryl)buta-2,3-dienoate (10.0 mmol, 1.0 equiv) in dry dichloromethane (20 mL) at −5 °C was added pyridine (15.0 mmol, 1.5 equiv), and then the mixture was stirred for 5.0 min at the same temperature. After that, acetyl chloride (15.0 mmol, 1.5 equiv) was slowly added into the mixture over 5 min and the contents were

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Scheme 8. Proposed Pathways for [3 + 3] and [4 + 1] Annulations



Scheme 9. Scale-Up Experiments for the Synthesis of 3aa, 4aa, and 6aa



stirred for 45 min. After completion of the reaction (TLC), the aqueous layer was extracted with dichloromethane (3 × 25 mL). Then the combined organic layer was washed with brine (2 × 20 mL), dried over anhydrous Na_2SO_4, and concentrated under reduced pressure. The crude product was purified by silica gel column chromatography using ethyl acetate/hexane (1:9) as the eluent.

*Benzyl 2-(Acetoxy(phenyl)methyl)buta-2,3-dienoate (**5a**).* Following the general procedure, the reaction of benzyl 2-(hydroxy(phenyl)methyl)buta-2,3-dienoate (2.80 g) with acetyl chloride (1.10 mL) and pyridine (1.20 mL) afforded **5a** as a gummy liquid using ethyl acetate/hexane (1:9) as the eluent. Yield: 1.84 g (57%). ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.31−7.28 (m, 2H), 7.23−7.21 (m, 5H), 7.19−7.16 (m, 3H), 6.53−6.52 (m, 1H), 5.19 (d, J = 2.5 Hz, 1H), 5.17 (d, J = 2.5 Hz, 1H), 5.06 (d, J = 3.5 Hz, 2H), 1.95 (s, 3H). ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 213.8, 169.6, 164.7, 138.1, 135.7, 128.5, 128.4$_1$, 128.3$_8$, 128.2, 128.0, 127.4, 102.6, 82.3, 71.7, 66.8, 21.0. IR (neat): ν_{max}3019, 2400, 1968, 1738, 1713, 1601, 1523, 1499, 1461, 1424, 1372, 1215, 1090, 1025, 979, 929, 853, 759, 698,

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669 cm^{-1}. HRMS (ESI-TOF): m/z [M + Na]$^+$ calcd for $C_{20}H_{18}NaO_4$345.1097, found 345.1091.

*Benzyl 2-(Acetoxy(4-methoxyphenyl)methyl)buta-2,3-dienoate (**5b**).* Following the general procedure, the reaction of benzyl 2-(hydroxy(4-methoxyphenyl)methyl)buta-2,3-dienoate (3.10 g) with acetyl chloride (1.10 mL) and pyridine (1.20 mL) afforded **5b** as a gummy liquid using ethyl acetate/hexane (1:9) as the eluent. Yield: 1.86 g (53%). ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.37−7.32 (m, 5H), 7.31−7.28 (m, 2H), 6.88 (d, J = 8.5 Hz, 2H), 6.59−6.58 (m, 1H), 5.35−5.33 (m, 2H), 5.19−5.17 (m, 2H), 3.82 (s, 3H), 2.07 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ 213.5, 169.7, 164.8, 159.7, 135.7, 130.1, 128.9, 128.5, 128.2, 128.0, 113.8, 102.7, 82.3, 71.3, 66.8, 55.3, 21.1. IR (neat): ν_{max}3019, 2961, 2839, 2400, 2254, 1969, 1713, 1612, 1514, 1462, 1424, 1373, 1216, 1176, 1091, 1029, 910, 854, 770, 669, 650 cm^{-1}. HRMS (ESI-TOF): m/z [M + Na]$^+$ calcd for $C_{21}H_{20}NaO_5$375.1203, found 375.1204.

*Benzyl 2-(Acetoxy(4-chlorophenyl)methyl)buta-2,3-dienoate (**5c**).* Following the general procedure, the reaction of benzyl 2-((4-chlorophenyl)(hydroxy)methyl)buta-2,3-dienoate (3.14 g) with acetyl chloride (1.10 mL) and pyridine (1.20 mL) afforded **5c** as a gummy liquid using ethyl acetate/hexane (1:9) as the eluent. Yield: 1.99 g (56%). ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.36−7.30 (m, 9H), 6.60−6.59 (m, 1H), 5.35 (d, J = 2.5 Hz, 1H), 5.32 (d, J = 2.5 Hz, 1H), 5.19 (d, J = 4.5 Hz, 2H), 2.09 (s, 3H). ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 213.6, 169.4, 164.6, 136.7, 135.6, 134.2, 128.7, 128.6, 128.5, 128.2, 128.0, 102.3, 82.4, 71.0, 66.9, 21.0. IR (neat): ν_{max}3054, 2986, 2308, 1742, 1712, 1498, 1425, 1265, 1090, 1026, 896, 743 cm^{-1}. HRMS (ESI-TOF): m/z [M + H]$^+$ calcd for $C_{20}H_{18}ClO_4$357.0888, found 357.0875.

*Benzyl 2-(Acetoxy(4-(trifluoromethyl)phenyl)methyl)buta-2,3-dienoate (**5d**).* Following the general procedure, the reaction of benzyl 2-(hydroxy(4-(trifluoromethyl)phenyl)methyl)buta-2,3-dienoate (3.48 g) with acetyl chloride (1.10 mL) and pyridine (1.20 mL) afforded **5d** as a gummy liquid using ethyl acetate/hexane (1:9) as the eluent. Yield: 2.30 g (60%). ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.62−7.60 (m, 2H), 7.53−7.52 (m, 2H), 7.37−7.34 (m, 3H), 7.31−7.30 (m, 2H), 6.68 (s, 1H), 5.36−5.35(m, 1H), 5.33 (d, J = 1.5 Hz, 1H), 5.21−5.20 (m, 2H), 2.11 (s, 3H). ^{19}F NMR (470 MHz, $CDCl_3$): δ −62.6; ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 213.8, 169.4, 164.5, 142.1, 135.5, 130.4 (q, $^2J_{C-F}$ = 32.4 Hz), 128.5, 128.3, 128.0, 127.5, 125.4 (q, $^3J_{C-F}$ = 3.6 Hz), 124.0 (q, $^1J_{C-F}$ = 270.3 Hz), 102.1, 82.6, 71.0, 67.0, 20.9. IR (neat): ν_{max}3019, 2400, 2254, 1969, 1742, 1712, 1422, 1374, 1326, 1216, 1170, 1131, 1067, 1019, 909, 856, 760, 668, 650 cm^{-1}. HRMS (ESI-TOF): m/z [M + Na]$^+$ calcd for $C_{21}H_{17}F_3NaO_4$413.0971, found 413.0970.

*Benzyl 2-(Acetoxy(pyren-4-yl)methyl)buta-2,3-dienoate (**5e**).* Following the general procedure, the reaction of benzyl 2-(hydroxy-(pyren-1-yl)methyl)buta-2,3-dienoate (4.04 g) with acetyl chloride (1.10 mL) and pyridine (1.20 mL) afforded **5e** as a brown solid using ethyl acetate/hexane (1:9) as the eluent. Yield: 2.0 g (45%), mp 114−116 °C. ^{1}H NMR (500 MHz, $CDCl_3$): δ 8.46−8.43 (m, 1H), 8.23−8.19 (m, 2H), 8.19−8.03 (m, 7H), 7.76−7.74 (m, 1H), 7.28−7.27 (m, 4H), 5.28−5.5.24 (m, 1H), 5.22−5.21 (m 2H), 5.14−5.11 (m, 1H), 2.15 (s, 3H). ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 214.3, 169.8, 164.9, 135.6, 131.4, 131.3, 131.1, 130.6, 128.4, 128.2, 128.1, 128.0, 127.8, 127.4, 126.1, 125.5, 125.4, 125.1, 124.9, 124.7$_4$, 124.6$_7$, 123.0, 102.8, 82.4, 69.5, 66.9, 21.1. IR (neat): ν_{max}3019, 2400, 2253, 1970, 1714, 1611, 1528, 1477, 1423, 1374, 1216, 1091, 1022, 909, 850, 760, 669, 650 cm^{-1}. HRMS (ESI-TOF): m/z [M + NH$_4$]$^+$ calcd for $C_{30}H_{26}NO_4$ 464.1856, found 464.1861.

*Benzyl 2-(Acetoxy(thiophene-2-yl)methyl)buta-2,3-dienoate (**5f**).* Following the general procedure, the reaction of benzyl 2-(hydroxy-(thiophene-2-yl)methyl)buta-2,3-dienoate (2.86 g) with acetyl chloride (1.10 mL) and pyridine (1.20 mL) afforded **5f** as a gummy liquid using ethyl acetate/hexane (1:9) as the eluent. Yield: 1.87 g (57%). ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.38−7.28 (m, 6H), 7.13 (d, J = 3.0 Hz, 1H), 6.99−6.97 (m, 1H), 6.91 (s, 1H), 5.41 (s, 2H), 5.22−5.21 (m, 2H), 2.09 (s, 3H). ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 213.3, 169.4, 164.5, 141.0, 135.6, 128.5, 128.2, 128.0, 126.9, 126.6, 126.0, 102.6, 82.9, 67.3, 66.9, 21.0. IR (neat): ν_{max}3019, 2400, 2254, 1969, 1739, 1713, 1601, 1531, 1424, 1373, 1216, 1092, 1022, 909, 855, 769, 669, 650 cm^{-1}. HRMS (ESI-TOF): m/z [M + H]$^+$ calcd for $C_{18}H_{17}NaO_4S$ 351.0662, found: 351.0666.

*Benzyl 3-Acetoxy-5-phenyl-2-vinylidenepentanoate (**5g**).* Following the general procedure, the reaction of benzyl 3-hydroxy-5-phenyl-2-vinylidenepentanoate (3.08 g) with acetyl chloride (1.10 mL) and pyridine (1.20 mL) afforded **5g** as a gummy liquid using ethyl acetate/hexane (1:9) as the eluent. Yield: 1.72 g (49%). ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.41−7.33 (m, 5H), 7.31−7.28 (m, 2H), 7.23−7.18 (m, 3H), 5.70−5.67 (m, 1H), 5.37−5.36 (m, 2H), 5.27−5.21 (m, 2H), 2.76−2.65 (m, 2H), 2.18−2.10 (m, 2H), 2.05 (s, 3H). ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 213.5, 170.0, 165.0, 141.2, 135.8, 128.5, 128.4, 128.3, 128.2, 128.0, 126.0, 101.6, 81.8, 69.9, 66.8, 35.1, 31.8, 21.0. IR (neat): 3027, 2927, 2856, 2237, 1963, 1739, 1713, 1496, 1372, 1229, 1102, 1029, 752, 698 cm^{-1}. HRMS (ESI-TOF): m/z [M + NH$_4$]$^+$ calcd for $C_{22}H_{26}NO_4$ 368.1856, found: 368.1853.

*iv. General Procedure for the Synthesis of Compounds **3aa**−**3aj** and **3ba**−**3da** and **3ea**−**3fa**.* A Schlenk tube was charged with 2-sulfonamido indole **1** (0.20 mmol), δ-acetoxy allenoate **2** (0.24 mmol), and toluene (2.0 mL). Subsequently, Ph_3P (0.04 mmol) was added to the reaction mixture at rt (25 °C), the mixture was stirred at the same temperature for 36 h, and the progress of the reaction was monitored using TLC. After completion of the reaction, the mixture was quenched by addition of water (10 mL). The aqueous layer was extracted with ethyl acetate (3 × 10 mL). Then the combined organic layer was washed with brine solution (20.0 mL), dried over anhydrous Na_2SO_4, and concentrated under reduced pressure. The crude product was then purified by silica gel column chromatography using ethyl acetate/hexane (1:9) as the eluent.

*Ethyl 2-(9-Ethyl-4-phenyl-1-tosyl-2,9-dihydro-1H-pyrido[2,3-b]indol-2-yl)acetate (**3aa**).* Following the general procedure, the reaction of **1a** (63.0 mg, 0.20 mmol) with **2a** (62.4 mg, 0.24 mmol) and PPh_3 (10.5 mg, 0.04 mmol) afforded **3aa** as a white solid. Yield: 85.4 mg (83%), mp 148−150 °C. ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.46 (d, J = 8.5 Hz, 1H), 7.39 (d, J = 8.5 Hz, 2H), 7.30−7.27 (m, 1H), 7.27−7.21 (m, 3H), 6.96−6.92 (m, 5H), 6.82 (d, J = 8.0 Hz, 1H), 5.41−5.37 (m, 1H), 5.14 (d, J = 6.0 Hz, 1H), 4.72−4.65 (m, 1H), 4.45−4.38 (m, 1H), 4.26−4.18 (m, 2H), 2.48−2.38 (m, 2H), 2.13 (s, 3H), 1.49 (t, J = 7.0 Hz, 3H), 1.31 (t, J = 7.0 Hz, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ 169.9, 144.3, 137.7, 136.1, 135.2, 134.0, 131.7, 129.0, 128.2, 127.9, 127.7, 122.9, 122.1, 120.7, 120.1, 116.1, 110.7, 105.6, 60.9, 55.4, 39.1, 37.9, 21.3, 14.7, 14.3. IR (neat): ν_{max} 3054, 2987, 1712, 1422, 1362, 1223, 1167, 896, 737, 706, 666 cm^{-1}. HRMS (ESI-TOF): m/z [M + H]$^+$ calcd for $C_{30}H_{31}N_2O_4S$ 515.1999, found 515.2006.

*Ethyl 2-(9-Ethyl-4-(4-methoxyphenyl)-1-tosyl-2,9-dihydro-1H-pyrido[2,3-b]indol-2-yl)acetate (**3ab**).* Following the general procedure, the reaction of **1a** (63.0 mg, 0.20 mmol) with **2b** (70.0 mg, 0.24 mmol) and PPh_3 (10.5 mg, 0.04 mmol) afforded **3ab** as a white solid. Yield: 92.5 mg (85%), mp 155−157 °C. ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.48 (d, J = 8.5 Hz, 1H), 7.41 (d, J = 8.0 Hz, 2H), 7.26−7.23 (m, 1H), 6.98−6.96 (m, 3H), 6.91−6.88 (m, 3H), 6.80 (d, J = 8.5 Hz, 2H), 5.39 (q, J = 7.5 Hz, 1H), 5.08 (d, J = 5.5 Hz, 1H), 4.73−4.66 (m, 1H), 4.46−4.39 (m, 1H), 4.27−4.17 (m, 2H), 3.85 (s, 3H), 2.48−2.38 (m, 2H), 2.17 (s, 3H), 1.51 (t, J = 7.0 Hz, 3H), 1.32 (t, J = 7.5 Hz, 3H). ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 169.9, 159.3, 144.2, 135.6, 135.2, 134.1, 131.7, 129.1, 129.0, 128.8, 128.2, 126.1, 123.0, 122.0, 120.7, 120.0, 115.0, 113.1, 110.7, 105.8, 60.8, 55.3, 39.1, 38.0, 21.3, 14.7, 14.2. IR (neat): ν_{max}3054, 2986, 1712, 1422, 1360, 1265, 1163, 896, 737, 706, 666 cm^{-1}. HRMS (ESI-TOF): m/z [M + H]$^+$ calcd for $C_{31}H_{33}N_2O_5S$ 545.2105, found 545.2104.

*Ethyl 2-(4-(4-Chlorophenyl)-9-ethyl-1-tosyl-2,9-dihydro-1H-pyrido[2,3-b]indol-2-yl)acetate (**3ac**).* Following the general procedure, the reaction of **1a** (63.0 mg, 0.20 mmol) with **2c** (70.7 mg, 0.24 mmol) and PPh_3 (10.5 mg, 0.04 mmol) afforded **3ac** as a yellow solid. Yield: 85.5 mg (78%), mp 142−144 °C. ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.49 (d, J = 8.5 Hz, 1H), 7.41 (d, J = 8.0 Hz, 2H), 7.26−7.23 (m, 3H), 7.01−6.97 (m, 3H), 6.92 (d, J = 8.0 Hz, 2H), 6.84 (d, J = 8.0 Hz, 1H), 5.43−5.39 (m, 1H), 5.16 (d, J = 5.5 Hz, 1H), 4.73−4.66 (m, 1H), 4.45−4.41 (m, 1H), 4.25−4.18 (m, 2H), 2.49−2.38

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(m, 2H), 2.18 (s, 3H), 1.51 (t, J = 7.0 Hz, 3H), 1.32 (t, J = 7.5 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 169.8, 144.3, 136.1, 135.2, 135.0, 134.0, 133.6, 131.9, 129.2, 129.0, 128.2, 127.9, 122.7, 122.2, 120.4, 120.3, 116.6, 110.8, 105.1, 60.9, 55.2, 39.2, 37.8, 21.4, 14.7, 14.3. IR (neat): ν_{max} 3054, 2986, 1731, 1598, 1535, 1492, 1440, 1422, 1360, 1265, 1168, 1090, 1016, 896, 738, 705, 665 cm^{-1}. HRMS (ESI-EXACTIVE ORBITRAP): m/z $[M + H]^+$ calcd for $C_{30}H_{30}ClN_2O_4S$ 549.1609, found 549.1634.

*Ethyl 2-(9-Ethyl-1-tosyl-4-(4-(trifluoromethyl)phenyl)-2,9-dihydro-1H-pyrido[2,3-b]indol-2-yl)acetate (**3ad**).* Following the general procedure, the reaction of **1a** (63.0 mg, 0.20 mmol) with **2d** (79.0 mg, 0.24 mmol) and PPh_3 (10.5 mg, 0.04 mmol) afforded **3ad** as a white solid. Yield: 90.0 mg (77%), mp 156−158 °C. ^1H NMR (500 MHz, $CDCl_3$): δ 7.53−7.50 (m, 3H), 7.41 (d, J = 8.5 Hz, 2H), 7.30−7.26 (m, 1H), 7.11 (d, J = 8.0 Hz, 2H), 7.02−6.99 (m, 1H), 6.97 (d, J = 8.0 Hz, 2H), 6.81 (d, J = 8.0 Hz, 1H), 5.45−5.41 (m, 1H), 5.23 (d, J = 5.5 Hz, 1H), 4.75−4.67 (m, 1H), 4.48−4.41 (m, 1H), 4.28−4.18 (m, 2H), 2.51−2.41 (m, 2H), 2.14 (s, 3H), 1.52 (t, J = 7.0 Hz, 3H), 1.33 (t, J = 7.0 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 169.9, 144.5, 141.3, 135.3, 135.0, 134.1, 132.1, 130.0 (q, $^2J_{C-F}$ = 32.0 Hz), 129.2, 128.3, 128.2, 124.8 (q, $^3J_{C-F}$ = 3.5 Hz), 124.3 (q, $^1J_{C-F}$ = 270.5 Hz), 122.7, 122.5, 120.5, 120.4, 118.0, 111.0, 104.9, 61.1, 55.3, 39.4, 37.9, 21.4, 14.8, 14.4; ^{19}F NMR (470 MHz, $CDCl_3$): −62.5. IR (neat): ν_{max}3054, 2986, 1732, 1714, 1611, 1535, 1441, 1422, 1361, 1325, 1265, 1168, 1126, 1089, 1068, 1018, 896, 851, 744, 706, 666 cm^{-1}. HRMS (ESI-EXACTIVE ORBITRAP): m/z $[M + H]^+$ calcd for $C_{31}H_{30}F_3N_2O_4S$ 583.1873, found 583.1894.

*Ethyl 2-(9-Ethyl-4-(3-methoxyphenyl)-1-tosyl-2,9-dihydro-1H-pyrido[2,3-b]indol-2-yl)acetate (**3ae**).* Following the general procedure, the reaction of **1a** (63.0 mg, 0.20 mmol) with **2e** (70.0 mg, 0.24 mmol) and PPh_3 (10.5 mg, 0.04 mmol) afforded **3ae** as a yellow solid. Yield: 82.7 mg (76%), mp 153−155 °C. ^1H NMR (500 MHz, $CDCl_3$): δ 7.48 (d, J = 8.5 Hz, 1H), 7.42 (d, J = 7.5 Hz, 2H), 7.26−7.23 (m, 1H), 7.20−7.17 (m, 1H), 7.00−6.96 (m, 3H), 6.90 (d, J = 8.0 Hz, 1H), 6.86−6.84 (m, 1H), 6.56 (d, J = 7.5 Hz, 1H), 6.50 (s, 1H), 5.44−5.40 (m, 1H), 5.19−5.18 (m, 1H), 4.74−4.66 (m, 1H), 4.47−4.40 (m, 1H), 4.28−4.18 (m, 2H), 3.78 (s, 3H), 2.50−2.40 (m, 2H), 2.18 (s, 3H), 1.51 (t, J = 7.0 Hz, 3H), 1.34−1.32 (m, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 170.0, 159.2, 144.5, 139.3, 136.1, 135.3, 134.1, 131.7, 129.2, 128.9, 128.8, 128.3, 123.0, 122.2, 120.8, 120.6, 120.2, 116.3, 113.9, 113.1, 110.8, 61.0, 55.5, 55.3, 39.3, 38.0, 21.4, 14.8, 14.4. IR (neat): ν_{max}3054, 2986, 1738, 1712, 1422, 1360, 1265, 1167, 896, 739, 706, 666 cm^{-1}. HRMS (ESI-EXACTIVE ORBITRAP): m/z $[M + H]^+$ calcd for $C_{31}H_{33}N_2O_5S$ 545.2105, found 545.2126.

*Ethyl 2-(4-(4-(Benzyloxy)-3-methoxyphenyl)-9-ethyl-1-tosyl-2,9-dihydro-1H-pyrido[2,3-b]indol-2-yl)acetate (**3af**).* Following the general procedure, the reaction of **1a** (63.0 mg, 0.20 mmol) with **2f** (95.1 mg, 0.24 mmol) and PPh_3 (10.5 mg, 0.04 mmol) afforded **3af** as a yellow solid. Yield: 97.5 mg (75%), mp 132−134 °C. ^1H NMR (500 MHz, $CDCl_3$): δ 7.49−7.48 (m, 3H), 7.43−7.41 (m, 4H), 7.36−7.33 (m, 1H), 7.27−7.24 (m, 1H), 7.00−6.92 (m, 4H), 6.79−6.77 (m, 1H), 6.55 (s, 1H), 6.48 (d, J = 8.0 Hz, 1H), 5.41−5.38 (m, 1H), 5.23−5.17 (m, 2H), 5.13−5.12 (m, 1H), 4.71−4.65 (m, 1H), 4.45−4.40 (m, 1H), 4.27−4.18 (m, 2H), 3.81 (s, 3H), 2.48−2.38 (m, 2H), 2.10 (s, 3H), 1.51 (t, J = 6.5 Hz, 3H), 1.32 (t, J = 7.0 Hz, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 170.0, 148.8, 147.8, 144.2, 137.1, 135.7, 135.2, 134.1, 131.7, 131.0, 129.0, 128.6, 128.2, 128.0, 127.4, 122.9, 122.1, 120.8, 120.4, 120.1, 115.1, 113.2, 111.7, 110.7, 105.6, 71.0, 60.9, 56.0, 55.4, 39.1, 38.0, 21.4, 14.7, 14.3. IR (neat): ν_{max}3054, 2986, 1731, 1712, 1600, 1511, 1439, 1422, 1360, 1265, 1167, 1089, 1023, 896, 730 cm^{-1}. HRMS (ESI-EXACTIVE ORBITRAP): m/z $[M + H]^+$ calcd for $C_{38}H_{39}N_2O_6S$ 651.2523, found 651.2550.

*Ethyl 2-(9-Ethyl-4-(naphthalen-1-yl)-1-tosyl-2,9-dihydro-1H-pyrido[2,3-b]indol-2-yl)acetate (**3ag**).* Following the general procedure, the reaction of **1a** (63.0 mg, 0.20 mmol) with **2g** (74.48 mg, 0.24 mmol) and PPh_3 (10.5 mg, 0.04 mmol) afforded **3ag** as a yellow solid. Yield: 81.2 mg (72%), mp 141−143 °C. ^1H NMR (500 MHz, $CDCl_3$): δ 7.87−7.85 (m, 1H), 7.77−7.72 (m, 2H), 7.52−7.50 (m, 4H), 7.44 (d, J = 8.0 Hz, 2H), 7.27−7.24 (m, 1H), 7.07 (d, J = 8.5 Hz, 1H), 6.97−6.90 (m, 3H), 6.84 (d, J = 8.0 Hz, 1H), 5.49−5.45 (m, 1H), 5.29 (d, J = 5.5 Hz, 1H), 4.76−4.69 (m, 1H), 4.50−4.43 (m, 1H), 4.31−4.24 (m, 2H), 2.55−2.46 (m, 2H), 2.00 (s, 3H), 1.54 (t, J = 7.0 Hz, 3H), 1.35 (t, J = 7.5 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 170.0, 144.5, 136.2, 135.3$_3$, 135.3$_1$, 134.2, 133.1$_4$, 133.0$_9$, 132.0, 129.1, 128.4, 128.1, 127.8, 127.3, 126.9, 126.3, 126.1, 123.1, 122.3, 120.9, 120.3, 116.9, 110.9, 105.8, 61.1, 55.6, 39.3, 38.1, 21.4, 14.9, 14.4. IR (neat): ν_{max}3054, 2986, 1712, 1606, 1422, 1362, 1265, 1167, 896, 706, 666 cm^{-1}. HRMS (ESIEXACTIVE ORBITRAP): m/z $[M + H]^+$ calcd for $C_{34}H_{33}N_2O_4S$ 565.2156, found 565.2179.

*Ethyl 2-(9-Ethyl-4-(thiophene-2-yl)-1-tosyl-2,9-dihydro-1H-pyrido[2,3-b]indol-2-yl)acetate (**3ah**).* Following the general procedure, the reaction of **1a** (63.0 mg, 0.20 mmol) with **2h** (63.91 mg, 0.24 mmol) and PPh_3 (10.5 mg, 0.04 mmol) afforded **3ah** as a brown solid. Yield: 76.0 mg (73%), mp 135−137 °C. ^1H NMR (500 MHz, $CDCl_3$): δ 7.49 (d, J = 8.0 Hz, 1H), 7.40 (d, J = 8.5 Hz, 2H), 7.30−7.26 (m, 1H), 7.20−7.18 m, 2H), 7.05−7.03 (m, 1H), 6.99−6.97 (m, 3H), 6.75−6.74 (m, 1H), 5.36−5.32 (m, 1H), 5.19 (d, J = 6.0 Hz, 1H), 4.71−4.64 (m, 1H), 4.46−4.38 (m, 1H), 4.27−4.18 (m, 2H), 2.47−2.36 (m, 2H), 2.17 (s, 3H), 1.51 (t, J = 7.0 Hz, 3H), 1.32 (t, J = 7.5 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 169.8, 144.4, 140.0, 135.1, 133.8, 131.7, 129.3, 129.1, 128.0, 126.7, 126.6, 124.6, 122.7, 122.2, 120.7, 120.2, 116.8, 110.7, 105.2, 60.9, 55.1, 39.1, 37.6, 21.4, 14.7, 14.2. IR (neat): ν_{max}3054, 2986, 1712, 1597, 1422, 1361, 1265, 1168, 1085, 896, 739, 705, 665 cm^{-1}. HRMS (ESI-EXACTIVE ORBITRAP): m/z $[M + H]^+$ calcd for $C_{28}H_{29}N_2O_4S_2$521.1563, found 521.1581.

*tert-Butyl 2-(2-(2-Ethoxy-2-oxoethyl)-9-ethyl-1-tosyl-2,9-dihydro-1H-pyrido[2,3-b]indol-4-yl)-1H-indole-1-carboxylate (**3ai**).* Following the general procedure, the reaction of **1a** (63.0 mg, 0.20 mmol) with **2i** (95.5 mg, 0.24 mmol) and PPh_3 (10.5 mg, 0.04 mmol) afforded **3ai** as a brown solid. Yield: 98.0 mg (75%), mp 114−116 °C. ^1H NMR (500 MHz, $CDCl_3$): δ 8.14 (d, J = 7.0 Hz, 1H), 7.48−7.45 (m, 3H), 7.32−7.29 (m, 1H), 7.26 (s, 1H), 7.24−7.20 (m, 1H), 7.10−7.07 (m, 1H), 7.01 (d, J = 8.0 Hz, 1H), 6.97 (d, J = 8.0 Hz, 2H), 6.90−6.87 (m, 2H), 5.47−5.42 (m, 1H), 5.38 (d, J = 5.5 Hz, 1H), 4.74−4.67 (m, 1H), 4.48−4.41 (m, 1H), 4.30−4.21 (m, 2H), 2.57−2.47 (m, 2H), 2.06 (s, 3H), 1.70 (s, 9H), 1.53 (t, J = 7.0 Hz, 3H), 1.34 (t, J = 7.5 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 169.9, 149.6, 144.3, 135.2, 135.1, 134.2, 131.4, 129.1, 128.9, 128.2, 127.3, 124.4, 124.3, 122.9, 122.5, 122.1, 120.9, 120.7, 120.2, 118.5, 116.5, 115.0, 110.6, 105.6, 84.0, 60.9, 55.3, 39.2, 38.2, 28.2, 21.3, 14.7, 14.3. IR (neat): ν_{max}3054, 2986, 2931, 1732, 1709, 1600, 1451, 1422, 1368, 1265, 1165, 1097, 1052, 896, 746, 706 cm^{-1}. HRMS (ESI-TOF): m/z $[M + H]^+$ calcd for $C_{37}H_{40}N_3O_6S$ 654.2632, found 654.2635.

*Ethyl 2-(9-Ethyl-4-methyl-1-tosyl-2,9-dihydro-1H-pyrido[2,3-b]-indol-2-yl)acetate (**3aj**).* Following the general procedure, the reaction of **1a** (63.0 mg, 0.20 mmol) with **2j** (47.5 mg, 0.24 mmol) and PPh_3 (10.5 mg, 0.04 mmol) afforded **3aj** as a gummy liquid. Yield: 64.2 mg (71%). ^1H NMR (500 MHz, $CDCl_3$): δ 7.64 (d, J = 8.0 Hz, 1H), 7.48 (d, J = 8.0 Hz, 1H), 7.31−7.27 (m, 3H), 7.17−7.14 (m, 1H), 7.08 (d, J = 8.0 Hz, 2H), 5.13−5.09 (m, 1H), 4.72−4.65 (m, 2H), 4.38−4.31 (m, 1H), 4.25−4.14 (m, 2H), 2.44−2.39 (m, 1H), 2.37 (s, 3H), 2.35−2.30 (m, 1H), 1.75 (s, 3H), 1.45 (t, J = 7.0 Hz, 3H), 1.30 (t, J = 7.0 Hz, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 170.1, 144.0, 135.0, 134.2, 130.7, 128.7, 128.2, 123.3, 121.9, 120.3, 120.0, 114.3, 110.9, 107.1, 60.8, 55.3, 39.1, 38.0, 21.5, 18.8, 14.6, 14.2. IR (neat): ν_{max} 2981, 1732, 1537, 1440, 1372, 1266, 1241, 1168, 1089, 1045, 733, 703 cm^{-1}. HRMS (ESI-TOF): m/z $[M + H]^+$ calcd for $C_{25}H_{29}N_2O_4S$ 453.1843, found 453.1846.

*Ethyl 2-(9-Methyl-4-phenyl-1-tosyl-2,9-dihydro-1H-pyrido[2,3-b]indol-2-yl)acetate (**3ba**).* Following the general procedure, the reaction of **1b** (60.0 mg, 0.20 mmol) with **2a** (62.4 mg, 0.24 mmol) and PPh_3 (10.5 mg, 0.04 mmol) afforded **3ba** as a yellow solid. Yield: 84.0 mg (84%), mp 125−127 °C. ^1H NMR (500 MHz, $CDCl_3$): δ 7.46−7.40 (m, 3H), 7.33−7.26 (m, 4H), 7.01−6.98 (m, 5H), 6.86 (d, J = 8.0 Hz, 1H), 5.39−5.35 (m, 1H), 5.11 (d, J = 5.5 Hz, 1H), 4.29−4.17 (m, 2H), 3.95 (s, 3H), 2.48−2.40 (m, 2H), 2.16 (s, 3H), 1.35 (t,

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J = 7.0 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 169.9, 144.3, 137.7, 136.2, 136.1, 134.1, 132.3, 129.1, 128.1, 127.9, 127.7, 122.4, 122.2, 120.5, 120.2, 115.7, 110.1, 105.2, 60.9, 55.5, 38.1, 31.1, 21.3, 14.3. IR (neat): ν_{max}3054, 2986, 1732, 1598, 1536, 1421, 1361, 1265, 1168, 1090, 1026, 896, 738, 705 cm^{-1}. HRMS (ESI-EXACTIVE ORBITRAP): m/z $[M + H]^+$ calcd for $C_{29}H_{29}N_2O_4S$ 501.1843, found 501.1861.

*Ethyl 2-(9-Allyl-4-phenyl-1-tosyl-2,9-dihydro-1H-pyrido[2,3-b]indol-2-yl)acetate (**3ca**).* Following the general procedure, the reaction of **1c** (65.3 mg, 0.20 mmol) with **2a** (62.4 mg, 0.24 mmol) and PPh_3 (10.5 mg, 0.04 mmol) afforded **3ca** as a yellow solid. Yield: 82.0 mg (78%), mp 124−126 °C. ^1H NMR (500 MHz, $CDCl_3$): δ 7.47 (d, J = 8.0 Hz, 1H), 7.42 (d, J = 7.5 Hz, 2H), 7.30−7.25 (m, 3H), 7.24−7.21 (m, 1H), 6.99−6.95 (m, 5H), 6.83 (d, J = 8.0 Hz, 1H), 6.12−6.04 (m, 1H), 5.42−5.38 (m, 1H), 5.29−5.17 (m, 4H), 5.06−5.02 (m, 1H), 4.27−4.16 (m, 2H), 2.49−2.40 (m, 2H), 2.16 (s, 3H), 1.33 (t, J = 7.0 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 169.9, 144.3, 137.6, 136.0, 135.9, 134.1, 133.9, 131.9, 129.1, 128.2, 127.9, 127.8, 127.7, 122.8, 122.2, 120.6, 120.3, 117.4, 116.2, 111.4, 105.7, 60.9, 55.3, 47.3, 38.0, 21.3, 14.3. IR (neat): ν_{max}3054, 2986, 1712, 1600, 1422, 1363, 1265, 1167, 896, 740, 705, 665 cm^{-1}. HRMS (ESI-EXACTIVE ORBITRAP): m/z $[M + H]^+$ calcd for $C_{31}H_{31}N_2O_4S$ 527.1999, found 527.2019.

*Ethyl 2-(9-Benzyl-4-phenyl-1-tosyl-2,9-dihydro-1H-pyrido[2,3-b]indol-2-yl)acetate (**3da**).* Following the general procedure, the reaction of **1d** (75.3 mg, 0.20 mmol) with **2a** (62.4 mg, 0.24 mmol) and PPh_3 (10.5 mg, 0.04 mmol) afforded **3da** as a white solid. Yield: 84.1 mg (73%), mp 124−126 °C. ^1H NMR (500 MHz, $CDCl_3$): δ 7.43 (d, J = 7.5 Hz, 2H), 7.27−7.23 (m, 8H), 7.13−7.09 (m, 4H), 6.99−6.91 (m, 5H), 6.83 (d, J = 7.5 Hz, 1H), 5.93−5.90 (m, 1H), 5.59−5.55 (m, 1H), 5.30 (d, J = 6.0 Hz, 1H), 5.14 (d, J = 5.0 Hz, 1H), 4.11−4.06 (m, 2H), 2.15 (s, 3H), 1.20 (t, J = 7.0 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 170.0, 144.4, 137.6, 137.5, 135.9, 135.8, 134.0, 132.3, 129.1, 128.5, 128.2, 128.0, 127.7$_3$, 127.7$_1$, 127.3, 127.0, 122.8, 122.4, 120.6, 120.3, 116.7, 111.4, 106.1, 60.7, 55.1, 48.0, 37.8, 21.3, 14.2. IR (neat): ν_{max} 3053, 3020, 2986, 1722, 1607, 1422, 1265, 1216, 1169, 909, 738, 668, 650 cm^{-1}. HRMS (ESI-TOF): m/z $[M + H]^+$ calcd for $C_{35}H_{33}N_2O_4S$ 577.2156, found 577.2157.

*Ethyl 2-(6-Bromo-9-ethyl-4-phenyl-1-tosyl-2,9-dihydro-1H-pyrido[2,3-b]indol-2-yl)acetate (**3ea**).* Following the general procedure, the reaction of **1e** (78.6 mg, 0.20 mmol) with **2a** (62.4 mg, 0.24 mmol) and PPh_3 (10.5 mg, 0.04 mmol) afforded **3ea** as a brown solid. Yield: 87.6 mg (74%), mp 160−162 °C. ^1H NMR (500 MHz, $CDCl_3$): δ 7.40−7.39 (m, 2H), 7.36−7.27 (m, 5H), 6.99 (d, J = 7.5 Hz, 2H), 6.95−6.92 (m, 3H), 5.43−5.39 (m, 1H), 5.18 (d, J = 5.5 Hz, 1H), 4.74−4.67 (m, 1H), 4.43−4.36 (m, 1H), 4.29−4.18 (m, 2H), 2.48−2.40 (m, 2H), 2.17 (s, 3H), 1.49 (t, J = 7.0 Hz, 3H), 1.33 (t, J = 7.5 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 169.7, 144.5, 137.1, 135.6, 133.9, 133.8, 132.6, 129.1, 128.0$_9$, 128.0$_5$, 127.9, 127.7, 125.0, 124.4, 123.1, 116.6, 113.5, 112.1, 105.2, 60.9, 55.4, 39.4, 38.0, 21.3, 14.6, 14.3. IR (neat): ν_{max}3054, 2986, 1712, 1603, 1422, 1361, 1265, 1169, 896, 743, 705, 666 cm^{-1}. HRMS (ESI-EXACTIVE ORBITRAP): m/z $[M + H]^+$ calcd for $C_{30}H_{30}BrN_2O_4S$ 593.1104, found 593.1126.

*Ethyl 2-(7-Bromo-9-ethyl-4-phenyl-1-tosyl-2,9-dihydro-1H-pyrido[2,3-b]indol-2-yl)acetate (**3fa**).* Following the general procedure, the reaction of **1f** (78.6 mg, 0.20 mmol) with **2a** (62.4 mg, 0.24 mmol) and PPh_3 (10.5 mg, 0.04 mmol) afforded **3fa** as a brown solid. Yield: 84.0 mg (71%), mp149−151 °C. ^1H NMR (500 MHz, $CDCl_3$): δ 7.63 (s, 1H), 7.40 (d, J = 8.5 Hz, 2H), 7.32−7.25 (m, 3H), 7.06−7.04 (m, 1H), 6.98 (d, J = 8.0 Hz, 2H), 6.93−6.92 (m, 2H), 6.68 (d, J = 8.5 Hz, 1H), 5.48−5.41 (m, 1H), 5.19 (d, J = 6.0 Hz, 1H), 4.73−4.65 (m, 1H), 4.41−4.34 (m, 1H), 4.28−4.20 (m, 2H), 2.49−2.40 (m, 2H), 2.16 (s, 3H), 1.50 (t, J = 7.0 Hz, 3H), 1.33 (t, J = 7.5 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 169.8, 144.5, 137.3, 135.9, 135.6, 133.8, 132.2, 129.1, 128.1, 127.9, 127.8$_1$, 127.8$_0$, 123.4, 121.9, 121.6, 116.6, 115.8, 113.6, 105.8, 61.0, 55.4, 39.3, 38.0, 21.3, 14.7, 14.3. IR (neat): ν_{max}3054, 2986, 1732, 1711, 1603, 1422, 1371, 1265, 1168, 1090, 1025, 896, 739, 705, 665 cm^{-1}. HRMS (ESI-EXACTIVE ORBITRAP): m/z $[M + H]^+$ calcd for $C_{30}H_{30}BrN_2O_4S$ 593.1104, found 593.1128.

*v. General Procedure for the Synthesis of Compounds **4aa**−**4ad** and **4af**, **4ah**, **4aj**, **4ak**, **4ba**, and **4ea**.* A Schlenk tube was charged with 2-sulfonamidoindole **1** (0.20 mmol), Ph_3P (0.04 mmol), cesium carbonate (0.40 mmol), and toluene (1.0 mL). Subsequently, δ-acetoxyallenoate **2** (0.24 mmol) in toluene (1.0 mL) was added gradually over 30 min at 80 °C, and the mixture was stirred at the same temperature for 36 h, and the progress of the reaction was monitored using TLC. After completion of the reaction, the mixture was quenched by addition of water (10 mL). The aqueous layer was extracted with ethyl acetate (3 × 15 mL). Then the combined organic layer was washed with brine solution (20 mL), dried over anhydrous Na_2SO_4, and concentrated under reduced pressure. The crude product was then purified by silica gel column chromatography using ethyl acetate/hexane (1:9) as the eluent.

*Ethyl 2-(9-Ethyl-4-phenyl-3-tosyl-9H-pyrido[2,3-b]indol-2-yl)-acetate (**4aa**).* Following the general procedure, the reaction of **1a** (63.0 mg, 0.20 mmol) with **2a** (62.4 mg, 0.24 mmol), PPh_3 (10.5 mg, 0.04 mmol), and Cs_2CO_3(130.0 mg, 0.40 mmol) afforded **4aa** as a white solid. Yield: 77.0 mg (75%), mp 162−164 °C. ^1H NMR (500 MHz, $CDCl_3$): δ 7.43−7.39 (m, 3H), 7.30−7.25 (m, 4H), 7.01 (d, J = 8.0 Hz, 2H), 6.96−6.94 (m, 2H), 6.91−6.87 (m, 1H), 6.23 (d, J = 8.0 Hz, 1H), 4.72 (s, 2H), 4.56 (q, J = 7.0 Hz, 2H), 4.28 (q, J = 7.0 Hz, 2H), 2.34 (s, 3H), 1.49 (t, J = 7.0 Hz, 3H), 1.34 (t, J = 7.0 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 171.9, 151.6, 150.7, 146.4, 142.9, 141.0, 140.1, 135.4, 129.3, 129.0, 128.0, 127.9, 127.3, 126.9, 126.5, 122.9, 120.7, 120.6, 115.3, 109.3, 60.9, 45.6, 36.4, 21.5, 14.3, 13.9. IR (neat): ν_{max}3054, 2985, 1712, 1563, 1495, 1470, 1420, 1366, 1311, 1265, 1223, 1190, 1150, 1092, 1032, 896, 739, 705, 666 cm^{-1}. HRMS (ESI-EXACTIVE ORBITRAP): m/z $[M + H]^+$ calcd for $C_{30}H_{29}N_2O_4S$ 513.1843, found 513.1852.

*Ethyl 2-(9-Ethyl-4-(4-methoxyphenyl)-3-tosyl-9H-pyrido[2,3-b]-indol-2-yl)acetate (**4ab**).* Following the general procedure, the reaction of **1a** (63.0 mg, 0.20 mmol) with **2b** (70.0 mg, 0.24 mmol), PPh_3 (10.5 mg, 0.04 mmol), and Cs_2CO_3(130.0 mg, 0.40 mmol) afforded **4ab** as a white solid. Yield: 85.0 mg (78%), mp 164−166 °C. ^1H NMR (500 MHz, $CDCl_3$): δ 7.43−7.42 (m, 2H), 7.30−7.28 (m, 2H), 7.03 (d, J = 8.0 Hz, 2H), 6.95−6.92 (m, 1H), 6.86 (d, J = 8.5 Hz, 2H), 6.80 (d, J = 8.5 Hz, 2H), 6.41 (d, J = 8.0 Hz, 1H), 4.72 (s, 2H), 4.56 (q, J = 7.0 Hz, 2H), 4.29 (q, J = 7.0 Hz, 2H), 3.93 (s, 3H), 2.35 (s, 3H), 1.50 (t, J = 7.0 Hz, 3H), 1.35 (t, J = 7.0 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 171.9, 159.4, 151.6, 150.7, 146.4, 142.7, 141.2, 140.1, 130.6, 128.9, 127.5, 127.2, 127.1, 126.8, 123.0, 120.7, 120.6, 115.7, 113.3, 109.3, 60.9, 55.4, 45.6, 36.4, 22.3, 14.3, 13.9. IR (neat): ν_{max}3054, 2986, 1712, 1422, 1365, 1265, 1152, 896, 743, 706, 666 cm^{-1}. HRMS (ESI-EXACTIVE ORBITRAP): m/z $[M + H]^+$ calcd for $C_{31}H_{31}N_2O_5S$ 543.1948, found 543.1963.

*Ethyl 2-(4-(4-Chlorophenyl)-9-ethyl-3-tosyl-9H-pyrido[2,3-b]-indol-2-yl)acetate (**4ac**).* Following the general procedure, the reaction of **1a** (63.0 mg, 0.20 mmol) with **2c** (71.0 mg, 0.24 mmol), PPh_3 (10.5 mg, 0.04 mmol), and Cs_2CO_3(130.0 mg, 0.40 mmol) afforded **4ac** as a white solid. Yield: 80.0 mg (73%); mp 189−191 °C. ^1H NMR (500 MHz, $CDCl_3$): δ 7.46−7.45 (m, 2H), 7.33 (d, J = 8.5 Hz, 2H), 7.28−7.25 (m, 2H), 7.07 (d, J = 8.5 Hz, 2H), 6.98−6.95 (m, 1H), 6.91 (d, J = 8.5 Hz, 2H), 6.38 (d, J = 8.0 Hz, 1H), 4.72 (s, 2H), 4.58 (q, J = 7.0 Hz, 2H), 4.29 (q, J = 7.0 Hz, 2H), 2.38 (s, 3H), 1.50 (t, J = 7.0 Hz, 3H), 1.35 (t, J = 7.0 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 171.9, 151.7, 150.7, 145.0, 143.2, 140.9, 140.2, 134.3, 133.8, 130.8, 129.0, 128.1, 127.5, 126.8, 126.6, 122.8, 120.8, 120.3, 115.1, 109.5, 60.9, 45.6, 36.4, 21.6, 14.3, 13.9. IR (neat): ν_{max}3054, 2986, 1712, 1612, 1561, 1422, 1265, 1164, 1092, 896, 742, 706, 666 cm^{-1}. HRMS (ESI-TOF): m/z $[M + H]^+$ calcd for $C_{30}H_{28}ClN_2O_4S$ 547.14539, found 547.1457.

*Ethyl 2-(9-Ethyl-3-tosyl-4-(4-(trifluoromethyl)phenyl)-9H-pyrido-[2,3-b]indol-2-yl)acetate (**4ad**).* Following the general procedure, the reaction of **1a** (63.0 mg, 0.20 mmol) with **2d** (79.0 mg, 0.24 mmol), PPh_3 (10.5 mg, 0.04 mmol), and Cs_2CO_3(130.0 mg, 0.40 mmol) afforded **4ad** as a white solid. Yield: 81.0 mg (70%); mp 138−140 °C. ^1H NMR (400 MHz, $CDCl_3$): δ 7.54 (d, J = 8.0 Hz, 2H), 7.48−7.46

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(m, 2H), 7.31−7.28 (m, 2H), 7.13 (d, J = 8.0 Hz, 2H), 7.04 (d, J = 8.0 Hz, 2H), 6.97−6.93 (m, 1H), 6.25 (d, J = 8.0 Hz, 1H), 4.73 (s, 2H), 4.60 (q, J = 7.2 Hz, 2H), 4.30 (q, J = 7.2 Hz, 2H), 2.38 (s, 3H), 1.52 (t, J = 7.2 Hz, 3H), 1.36 (t, J = 7.2 Hz, 3H). ^{19}F NMR (470 MHz, $CDCl_3$): δ −62.6; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 171.8, 151.7, 150.7, 144.5, 143.3, 140.8, 140.3, 139.3, 130.2 (q, $^2J_{C-F}$ = 32.3 Hz), 129.9, 129.1, 127.6, 126.8, 126.5, 124.7 (q, $^3J_{C-F}$ = 3.1 Hz), 124.1 (q, $^1J_{C-F}$ = 270.6 Hz), 122.5, 120.9, 120.2, 114.7, 109.6, 61.0, 45.5, 36.5, 21.5, 14.3, 13.9. IR (neat): ν_{max}3054, 2986, 1712, 1562, 1422, 1365, 1324, 1265, 1151, 1092, 1021, 896, 744, 706, 666 cm^{-1}. HRMS (ESI-TOF): m/z [M + H]$^+$ calcd for $C_{31}H_{28}F_3N_2O_4S$ 581.1716, found 581.1716.

*Ethyl 2-(4-(4-(Benzyloxy)-3-methoxyphenyl)-9-ethyl-3-tosyl-9H-pyrido[2,3-b]indol-2-yl)acetate (**4af**).* Following the general procedure, the reaction of **1a** (63.0 mg, 0.20 mmol) with **2f** (95.1 mg, 0.24 mmol), PPh_3 (10.5 mg, 0.04 mmol), and Cs_2CO_3(130.0 mg, 0.40 mmol) afforded **4af** as a white solid. Yield: 88.0 mg (68%); mp 176−178 °C. ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.57 (d, J = 7.5 Hz, 2H), 7.50−7.47 (m, 2H), 7.44−7.39 (m, 3H), 7.25 (d, J = 8.0 Hz, 2H), 6.92 (d, J = 8.0 Hz, 3H), 6.85 (d, J = 8.0 Hz, 1H), 6.56 (d, J = 6.5 Hz, 1H), 6.42 (d, J = 7.5 Hz, 1H), 6.29 (s, 1H), 5.36−5.28 (m, 2H), 4.85−4.81 (m, 1H), 4.65−4.62 (m, 1H), 4.59−4.55 (m, 2H), 4.33−4.28 (m, 2H), 3.57 (s, 3H), 2.30 (s, 3H), 1.50 (t, J = 7.0 Hz, 3H), 1.35 (t, J = 7.0 Hz, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 172.2, 151.4, 150.7, 149.1, 147.6, 146.1, 142.7, 141.1, 140.1, 137.0, 128.8, 128.7, 128.1, 127.8, 127.3, 127.2, 127.1, 123.1, 122.2, 120.7, 120.6, 115.6, 113.3, 112.6, 109.4, 70.9, 60.9, 55.4, 45.8, 36.4, 21.5, 14.3, 14.0. IR (neat): ν_{max} 3054, 2986, 2928, 2856, 1731, 1611, 1511, 1422, 1265, 1164, 896, 746, 666 cm^{-1}. HRMS (ESI-TOF): m/z [M + H]$^+$ calcd for $C_{38}H_{37}N_2O_6S$ 649.2367, found 649.2367.

*Ethyl 2-(9-Ethyl-4-(thiophene-2-yl)-3-tosyl-9H-pyrido[2,3-b]-indol-2-yl)acetate (**4ah**).* Following the general procedure, the reaction of **1a** (63.0 mg, 0.20 mmol) with **2h** (64.0 mg, 0.24 mmol), PPh_3 (10.5 mg, 0.04 mmol), and Cs_2CO_3 (130.0 mg, 0.40 mmol) afforded **4ah** as a white solid. Yield: 66.3 mg (64%); mp 148−150 °C. ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.48−7.42 (m, 5H), 7.10−7.07 (m, 3H), 7.03−6.99 (m, 1H), 6.83−6.82 (m, 1H), 6.42 (d, J = 8.0 Hz, 1H), 4.79−4.76 (m, 1H), 4.69−4.65 (m, 1H), 4.57 (q, J = 7.0 Hz, 2H), 4.31−4.27 (m, 2H), 2.37 (s, 3H), 1.50 (t, J = 7.0 Hz, 3H), 1.35 (t, J = 7.0 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 171.8, 151.7, 150.6, 142.9, 140.9, 140.3, 139.2, 134.9, 129.4, 129.1, 128.0, 127.9, 127.6, 126.8, 126.7, 123.2, 120.9, 120.3, 117.0, 109.4, 60.9, 45.6, 36.4, 21.6, 14.3, 13.9. IR (neat): ν_{max}3054, 2986, 1712, 1605, 1422, 1265, 1147, 896, 745, 706, 666 cm^{-1}. HRMS (ESI-TOF): m/z [M + H]$^+$ calcd for $C_{28}H_{27}N_2O_4S_2$ 519.1407, found 519.1408.

*Ethyl 2-(9-Ethyl-4-methyl-3-tosyl-9H-pyrido[2,3-b]indol-2-yl)-acetate (**4aj**).* Following the general procedure, the reaction of **1a** (63.0 mg, 0.20 mmol) with **2a** (47.5 mg, 0.24 mmol), PPh_3 (10.5 mg, 0.04 mmol), and Cs_2CO_3 (130 mg, 0.40 mmol) afforded **4aj** as a white solid. Yield: 61.2 mg (68%), mp 133−135 °C. ^{1}H NMR (500 MHz, $CDCl_3$): δ 8.11 (d, J = 8.0 Hz, 1H), 7.83 (d, J = 8.0 Hz, 2H), 7.59−7.52 (m, 2H), 7.34−7.28 (m, 3H), 4.64 (s, 2H), 4.56 (q, J = 7.0 Hz, 2H), 4.25 (q, J = 7.0 Hz, 2H), 2.96 (s, 3H), 2.41 (s, 3H), 1.48 (t, J = 7.0 Hz, 3H), 1.32 (t, J = 7.0 Hz, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 171.7, 152.3, 150.7, 145.0, 143.4, 141.4, 140.0, 129.5, 127.0, 126.6, 126.0, 123.6, 121.1, 120.9, 115.0, 109.7, 60.8, 45.5, 36.3, 21.6, 18.5, 14.3, 13.9. IR (neat): ν_{max} 2926, 1711, 1568, 1495, 1409, 1362, 1304, 1220, 1162, 1146, 1107, 1088, 1027, 816, 740, 706, 679 cm^{-1}. HRMS (ESI-TOF): m/z [M + H]$^+$ calcd for $C_{25}H_{27}N_2O_4S$ 451.1686, found 451.1687.

*Ethyl 2-(4-(2-(Benzyloxy)naphthalen-1-yl)-9-ethyl-3-tosyl-9H-pyrido[2,3-b]indol-2-yl)acetate (**4aj**).* Following the general procedure, the reaction of **1a** (63.0 mg, 0.20 mmol) with **2k** (100.0 mg, 0.24 mmol), PPh_3 (10.5 mg, 0.04 mmol), and Cs_2CO_3 (130.0 mg, 0.40 mmol) afforded **4ak** as a white solid. Yield: 96.2 mg (72%); mp 115−117 °C. ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.99 (d, J = 9.0 Hz, 1H), 7.78 (d, J = 8.0 Hz, 1H), 7.40 (d, J = 8.5 Hz, 1H), 7.35−7.30 (m, 2H), 7.23−7.20 (m, 3H), 7.15−7.08 (m, 3H), 6.96−6.93 (m, 3H), 6.77 (d, J = 8.5 Hz, 1H), 6.73−6.69 (m, 3H), 5.97 (d, J = 8.0 Hz, 1H), 5.02−4.99 (m, 1H), 4.94−4.91 (m, 1H), 4.86−4.82 (m, 1H), 4.61−4.56 (m, 3H), 4.29 (q, J = 7.0 Hz, 2H), 2.20 (s, 3H), 1.53 (t, J = 7.0 Hz, 3H), 1.34 (t, J = 7.0 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 171.6, 153.2, 152.2, 151.0, 142.7, 142.3, 140.2, 140.0, 137.3, 132.4, 130.7, 128.6, 128.5, 128.2, 127.7, 127.6, 127.3, 127.0, 126.7, 126.5, 124.5, 123.4, 122.5, 120.7, 120.6, 118.9, 115.4, 114.4, 109.2, 70.5, 60.8, 45.5, 36.5, 21.3, 14.3, 14.0. IR (neat): ν_{max} 3054, 2986, 1738, 1601, 1557, 1422, 1265, 1153, 896, 748, 666 cm^{-1}. HRMS (ESI-TOF): m/z [M + H]$^+$ calcd for $C_{41}H_{37}N_2O_5S$ 669.2418, found 669.2396.

*Ethyl 2-(9-Methyl-4-phenyl-3-tosyl-9H-pyrido[2,3-b]indol-2-yl)-acetate (**4ba**).* Following the general procedure, the reaction of **1b** (60. mg, 0.20 mmol) with **4a** (62.4 mg, 0.24 mmol), PPh_3 (10.5 mg, 0.04 mmol), and Cs_2CO_3 (130.0 mg, 0.40 mmol) afforded **4ba** as a white solid. Yield: 76.0 mg (76%); mp 130−132 °C. ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.46−7.41 (m, 3H), 7.30−7.26 (m, 4H), 7.01 (d, J = 8.5 Hz, 2H), 6.96−6.91 (m, 3H), 6.24 (d, J = 8.0 Hz, 1H), 4.73 (s, 2H), 4.30 (q, J = 7.0 Hz, 2H), 4.00 (s, 3H), 2.36 (s, 3H), 1.36 (t, J = 7.0 Hz, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 171.9, 151.6, 151.3, 146.5, 142.9, 141.2, 141.0, 135.4, 129.3, 129.0, 128.0, 127.9, 127.4, 126.9, 126.6, 122.8, 120.9, 120.4, 115.4, 109.2, 60.9, 45.6, 27.8, 21.5, 14.3. IR (neat): ν_{max}3004, 2966, 2925, 1716, 1421, 1362, 1222, 1092, 902, 787, 665 cm^{-1}. HRMS (ESI-TOF): m/z [M + H]$^+$ calcd for $C_{29}H_{27}N_2O_4S$ 499.1686, found 499.1687.

*Ethyl 2-(6-Bromo-9-ethyl-4-phenyl-3-tosyl-9H-pyrido[2,3-b]-indol-2-yl)acetate (**4ea**).* Following the general procedure, the reaction of **1e** (79.0 mg, 0.20 mmol) with **2a** (62.4 mg, 0.24 mmol), PPh_3 (10.5 mg, 0.04 mmol), and Cs_2CO_3 (130.0 mg, 0.40 mmol) afforded **4ea** as a white solid. Yield: 77.0 mg (65%); mp 202−204 °C. ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.51 (d, J = 8.5 Hz, 1H), 7.47−7.44 (m, 1H), 7.31−7.28 (m, 5H), 7.02 (d, J = 8.0 Hz, 2H), 6.93 (d, J = 7.5 Hz, 2H), 6.27 (s, 1H), 4.72 (s, 2H), 4.55 (q, J = 6.5 Hz, 2H), 4.30 (q, J = 7.0 Hz, 2H), 2.36 (s, 3H), 1.49 (t, J = 7.0. Hz, 3H), 1.35 (t, J = 7.0 Hz, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 171.9, 152.3, 150.7, 147.0, 143.1, 140.8, 138.8, 134.9, 130.0, 129.0, 128.3, 128.1, 127.2, 127.0, 125.7, 122.2, 114.6, 113.5, 110.8, 61.0, 45.6, 36.6, 21.6, 14.3, 13.9. IR (neat): ν_{max}3054, 2985, 2927, 1711, 1604, 1574, 1422, 1365, 1265, 1167, 1091, 1027, 976, 896, 858, 737, 666 cm^{-1}. HRMS (ESI-TOF): m/z [M + H]$^+$ calcd for $C_{30}H_{28}BrN_2O_4S$ 591.0948, found 591.0948.

*vi. General Procedure for the Synthesis of Compounds **6aa**−**6ag**, **6ba**−**6da**, **6bf**, **6ga**, and **6hb**.* A Schlenk tube was charged with 2-sulfonamido indole **1** (0.20 mmol), tri(p-tolyl)phosphine (0.04 mmol), and toluene (1.0 mL). Subsequently, β'-acetoxyallenoate **5** (0.24 mmol) in toluene (1.0 mL) was added gradually over 30 min at 80 °C, the mixture was stirred at the same temperature for the stipulated time (36 h), and the progress of the reaction was monitored using TLC. After completion of the reaction, the mixture was quenched by addition of water (10 mL). The aqueous layer was extracted with ethyl acetate (3 × 15 mL). Then the combined organic layer was washed with brine solution (20 mL), dried over anhydrous Na_2SO_4, and concentrated under reduced pressure. The crude product was then purified by silica gel column chromatography using ethyl acetate/hexane (1:4) as the eluent.

*Benzyl (Z)-1′-Ethyl-2-phenyl-2′-(tosylimino)spiro[cyclopentane-1,3′-indolin]-3-ene-3-carboxylate (**6aa**).* Following the general procedure, the reaction of **1a** (63.0 mg, 0.20 mmol) with **5a** (77.4 mg, 0.24 mmol) and tri(p-tolyl)phosphine (12.2 mg, 0.04 mmol) afforded **6aa** as a white solid. Yield: 91.0 mg (79%), mp 148−150 °C. ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.99−7.96 (m, 2H), 7.28−7.20 (m, 6H), 7.16−7.14 (m, 1H), 7.05−7.02 (m, 1H), 6.97−6.87 (m, 6H), 6.72−6.71 (m, 2H), 6.57 (d, J = 8.0 Hz, 1H), 5.65−5.63 (m, 1H), 5.20−5.17 (m, 1H), 5.01−4.98 (m, 1H), 4.19−4.15 (m, 1H), 4.08−4.01 (m, 1H), 3.75−3.68 (m, 1H), 2.97−2.92 (m, 1H), 2.43 (s, 3H), 1.20 (t, J = 7.0 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 170.2, 164.2, 144.3, 142.3, 141.0, 140.4, 136.8, 136.7, 135.7, 134.3, 129.3, 128.3, 128.1, 127.8, 127.7, 127.3, 126.6, 126.4, 124.1, 123.2, 108.6, 66.1, 62.3, 60.1, 42.8, 37.6, 21.5, 12.0. IR (neat): ν_{max}3054, 2986, 1712, 1603, 1565, 1473, 1422, 1265, 1165, 1086, 896, 705, 666 cm^{-1}. HRMS (ESI-TOF): m/z [M + H]$^+$ calcd for $C_{35}H_{33}N_2O_4S$ 577.2156, found 577.2162.

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*Benzyl (Z)-1′-Ethyl-2-(4-methoxyphenyl)-2′-(tosylimino)spiro-[cyclopentane-1,3′-indolin]-3-ene-3-carboxylate (**6ab**).* Following the general procedure, the reaction of **1a** (63.0 mg, 0.20 mmol) with **5b** (84.5 mg, 0.24 mmol) and tri(*p*-tolyl)phosphine (12.2 mg, 0.04 mmol) afforded **6ab** as a white solid. Yield: 100.6 mg (83%), mp 175−177 °C. ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.96 (d, J = 7.5 Hz, 2H), 7.28−7.24 (m, 6H), 7.15 (d, J = 7.0 Hz, 1H), 7.06−6.99 (m, 3H), 6.91−6.88 (m, 1H), 6.64−6.58 (m, 3H), 6.44 (d, J = 8.0 Hz, 2H), 5.58 (s, 1H), 5.21−5.19 (m, 1H), 4.99−4.97 (m, 1H), 4.15−4.12 (m, 1H), 4.05−4.01 (m, 1H), 3.74−3.70 (m, 1H), 3.64 (s, 3H), 2.94−2.90 (m, 1H), 2.43 (s, 3H), 1.20 (t, J = 6.0 Hz, 3H). ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 170.2, 164.2, 158.2, 143.9, 142.2, 141.1, 140.5, 137.1, 135.8, 134.4, 129.2$_3$, 129.1$_7$, 128.9, 128.2, 128.1, 127.8, 126.3, 124.0, 123.2, 112.8, 108.7, 66.0, 62.5, 59.5, 55.0, 42.6, 37.6, 21.5, 12.0. IR (neat): ν_{max}3054, 2986, 2925, 1712, 1556, 1422, 1363, 1265, 1149, 1087, 896, 745, 666 cm^{-1}. HRMS (ESI-TOF): m/z [M + H]$^+$ calcd for $C_{36}H_{35}N_2O_5S$ 607.2261, found 607.2263.

*Benzyl (Z)-2-(4-Chlorophenyl)-1′-ethyl-2′-(tosylimino)spiro-[cyclopentane-1,3′-indolin]-3-ene-3-carboxylate (**6ac**).* Following the general procedure, the reaction of **1a** (63.0 mg, 0.20 mmol) with **5c** (85.4 mg, 0.24 mmol) and tri(*p*-tolyl)phosphine (12.2 mg, 0.04 mmol) afforded **6ac** as a white solid. Yield: 92.7 mg (76%), mp 170−172 °C. ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.98−7.95 (m, 2H), 7.31−7.24 (m, 6H), 7.17 (d, J = 7.5 Hz, 1H), 7.10−7.06 (m, 1H), 7.01−6.99 (m, 2H), 6.93−6.90 (m, 1H), 6.86 (d, J = 8.0 Hz, 2H), 6.65 (d, J = 8.5 Hz, 2H), 6.61 (d, J = 8.0 Hz, 1H), 5.65 (d, J = 2.5 Hz, 1H), 5.24−5.21 (m, 1H), 4.97−4.95 (m, 1H), 4.24−4.19 (m, 1H), 4.04−3.97 (m, 1H), 3.74−3.67 (m, 1H), 2.95−2.91 (m, 1H), 2.44 (s, 3H), 1.19 (t, J = 7.0 Hz, 3H). ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 169.8, 163.9, 144.7, 142.4, 140.9, 140.3, 136.4, 135.6, 135.3 134.0, 132.3, 129.5, 129.3, 128.5, 128.3, 128.0, 127.9, 127.5, 126.3, 123.9, 123.3, 108.8, 62.2, 59.4, 42.7, 37.6, 21.5, 12.1. IR (neat): ν_{max}3054, 2986, 2928, 1712, 1624, 1560, 1490, 1422, 1376, 1265, 1148, 1087, 1046, 1015, 958, 896, 818, 739 cm^{-1}. HRMS (ESI-TOF): m/z [M + H]$^+$ calcd for $C_{35}H_{32}ClN_2O_4S$ 611.1766, found 611.1766.

*Benzyl (Z)-1′-Ethyl-2′-(tosylimino)-2-(4-(trifluoromethyl)phenyl)-spiro[cyclopentane-1,3′-indolin]-3-ene-3-carboxylate (**6ad**).* Following the general procedure, the reaction of **1a** (63.0 mg, 0.20 mmol) with **5d** (94.0 mg, 0.24 mmol) and tri(*p*-tolyl)phosphine (12.2 mg, 0.04 mmol) afforded **6ad** as a white solid. Yield: 91.5 mg (71%), mp 123−125 °C. ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.98 (d, J = 8.5 Hz, 2H), 7.35−7.31 (m, 3H), 7.27−7.21 (m, 3H), 7.18 (d, J = 7.5 Hz, 1H), 7.14 (d, J = 8.0 Hz, 2H), 7.07−7.05 (m, 1H), 6.98 (d, J = 7.0 Hz, 2H), 6.92−6.89 (m, 1H), 6.83 (d, J = 8.0 Hz, 2H), 6.58 (d, J = 8.0 Hz, 1H), 5.75 (d, J = 2.0 Hz, 1H), 5.25−5.23 (m, 1H), 4.96−4.94 (m, 1H), 4.29−4.24 (m, 1H), 4.03−3.96 (m, 1H), 3.74−3.67 (m, 1H), 2.98−2.93 (m, 1H), 2.45 (s, 3H), 1.18 (t, J = 7.0 Hz, 3H). ^{19}F NMR (376 MHz, $CDCl_3$): δ −62.5; ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 169.6, 163.8, 145.2, 142.5, 141.0, 140.8, 140.2, 136.0, 135.4, 133.7, 129.3, 128.3 (q, $^2J_{C-F}$ = 32.4 Hz), 128.5, 128.4, 128.3, 128.1, 128.0, 126.3, 124.1 (q, $^3J_{C-F}$ = 3.5 Hz), 124.0 (q, $^1J_{C-F}$ = 270.3 Hz), 123.8, 123.3, 108.8, 66.3, 62.2, 59.6, 42.8, 37.5, 21.5, 12.0. IR (neat): ν_{max}3054, 2986, 1712, 1561, 1422, 1363, 1326, 1265, 1223, 1087, 896, 740, 705, 665 cm^{-1}. HRMS (ESI-TOF): m/z [M + H]$^+$ calcd for $C_{36}H_{32}F_3N_2O_4S$ 645.2029, found 645.2029.

*Benzyl (Z)-1′-Ethyl-2-(pyren-2-yl)-2′-(tosylimino)spiro-[cyclopentane-1,3′-indolin]-3-ene-3-carboxylate (**6ae**).* Following the general procedure, the reaction of **1a** (63.0 mg, 0.20 mmol) with **5e** (107.1 mg, 0.24 mmol) and tri(*p*-tolyl)phosphine (12.2 mg, 0.04 mmol) afforded **6ae** as a white solid. Yield: 91.0 mg (65%), mp 194−196 °C. ^{1}H NMR (500 MHz, $CDCl_3$): δ 8.15−8.07 (m, 5H), 7.98−7.95 (m, 2H), 7.87 (d, J = 8.5 Hz, 1H), 7.77−7.52 (m, 2H), 7.59 (d, J = 8.0 Hz, 1H), 7.43−7.42 (m, 1H), 7.36 (d, J = 8.0 Hz, 2H), 7.10−7.09 (m, 1H), 6.97−6.94 (m, 1H), 6.77−6.68 (m, 5H), 6.59 (d, J = 7.5 Hz, 2H), 6.24 (d, J = 7.5 Hz, 1H), 5.11−5.08 (m, 1H), 4.88−4.85 (m, 1H), 4.25−4.21 (m, 1H), 3.78−3.63 (m, 2H), 3.15−3.11 (m, 1H), 2.48 (s, 3H), 0.99 (t, J = 7.0 Hz, 3H). ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 170.8, 164.2, 143.9, 142.4, 141.1, 140.4, 137.9, 135.5, 134.2, 131.8, 131.2, 130.6, 130.0, 129.4, 129.3, 127.9$_4$, 127.8$_9$, 127.5$_1$, 127.4$_5$, 127.3$_6$, 127.1, 126.9, 126.8, 126.5, 125.7, 124.9, 124.6, 124.5$_3$, 124.4$_7$, 123.9, 123.1, 123.0, 108.7, 66.0, 62.2, 55.8, 44.1, 38.0, 21.6, 11.4. IR (neat): ν_{max}3054, 2986, 1713, 1560, 1488, 1421, 1375, 1265, 1148, 1087, 1016, 959, 896, 841, 818, 739 cm^{-1}. HRMS (ESI-TOF): m/z [M + H]$^+$ calcd for $C_{45}H_{37}N_2O_4S$ 701.2469, found 701.2469.

*Benzyl (Z)-1′-Ethyl-2-(thiophene-2-yl)-2′-(tosylimino)spiro-[cyclopentane-1,3′-indolin]-3-ene-3-carboxylate (**6af**).* Following the general procedure, the reaction of **1a** (63.0 mg, 0.20 mmol) with **5f** (79.0 mg, 0.24 mmol) and tri(*p*-tolyl)phosphine (12.2 mg, 0.04 mmol) afforded **6af** as a brown solid. Yield: 81.5 mg (70%), mp 77−79 °C. ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.93 (d, J = 8.5 Hz, 2H), 7.28−7.25 (m, 5H), 7.23−7.22 (m, 1H), 7.16−7.13 (m, 2H), 7.09−7.07 (m, 2H), 6.95−6.92 (m, 1H), 6.88 (d, J = 5.0 Hz, 1H), 6.72−6.70 (m, 1H), 6.63−6.61 (m, 1H), 6.46−6.45 (m, 1H), 5.83 (d, J = 2.5 Hz, 1H), 5.22−5.20 (m, 1H), 5.05−5.03 (m, 1H), 4.10−4.03 (m, 2H), 3.85−3.78 (m, 1H), 2.98−2.94 (m, 1H), 2.42 (s, 3H), 1.25 (t, J = 7.0 Hz, 3H). ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 169.8, 163.8, 143.9, 142.3, 140.9$_2$, 140.8$_6$, 140.0, 137.0, 135.8, 133.9, 129.2, 128.4, 128.3, 127.9, 127.8, 126.3, 126.1, 125.9, 124.2, 124.0, 123.4, 108.8, 66.2, 62.1, 55.0, 43.1, 37.8, 21.5, 12.1. IR (neat): ν_{max}3054, 2986, 1713, 1632, 1561, 1488, 1422, 1379, 1265, 1148, 1087, 1014, 896, 819, 738, 705, 666 cm^{-1}. HRMS (ESI-TOF): m/z [M + H]$^+$ calcd for $C_{33}H_{31}N_2O_4S_2$583.1720, found 583.1722.

*Benzyl 1′-Ethyl-2-phenethyl-2′-(tosylimino)spiro[cyclopent[3]-ene-1,3′-indoline]-3-carboxylate (**6ag**).* Following the general procedure, the reaction of **1a** (63.0 mg, 0.20 mmol) with **5g** (84.0 mg, 0.24 mmol) and tri(*p*-tolyl)phosphine (12.2 mg, 0.04 mmol) afforded **6ag** as a gummy liquid. Yield: 83.4 mg (69%); ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.92 (d, J = 7.5 Hz, 2H), 7.41−7.36 (m, 7H), 7.29 (d, J = 6.5 Hz, 2H), 7.16−7.02 (m, 6H), 6.55 (d, J = 6.5 Hz, 2H), 5.29−5.26 (m, 1H), 5.19−5.16 (m, 1H), 4.54 (d, J = 11.0 Hz, 1H), 4.14−4.10 (m, 1H), 4.01−3.97 (m, 1H), 3.88−3.83 (m, 1H), 2.71−2.67 (m, 1H), 2.44−2.41 (m, 4H), 2.13−2.08 (m, 1H), 1.65−1.59 (m, 1H), 1.44−1.37 (m, 1H), 1.29 (t, J = 7.0 Hz, 3H). ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 171.1, 164.1, 142.3, 141.9, 141.8, 141.1, 140.9, 137.9, 135.9, 134.7, 129.2, 128.5$_8$, 128.5$_5$, 128.3, 128.2, 128.1, 128.0, 126.4, 125.6, 123.9, 123.8, 109.4, 66.3, 59.9, 52.8, 45.1, 37.6, 33.6, 31.7, 21.5, 11.6. IR (neat): ν_{max} 2923, 2853, 1710, 1556, 1487, 1468, 1373, 1336, 1282, 1236, 1172, 1146, 1084, 955, 900, 815, 772, 753, 698, 679 cm^{-1}. HRMS (ESI-TOF): m/z [M + H]$^+$ calcd for $C_{37}H_{37}N_2O_4S$ 605.2469, found 605.2463.

*Benzyl (Z)-1′-Methyl-2-phenyl-2′-(tosylimino)spiro-[cyclopentane-1,3′-indolin]-3-ene-3-carboxylate (**6ba**).* Following the general procedure, the reaction of **1b** (60.0 mg, 0.20 mmol) with **5a** (77.4 mg, 0.24 mmol) and tri(*p*-tolyl)phosphine (12.2 mg, 0.04 mmol) afforded **6ba** as a white solid. Yield: 92.2 mg (82%), mp 105−107 °C. ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.97 (d, J = 8.0 Hz, 2H), 7.29−7.22 (m, 6H), 7.08−6.95 (m, 7H), 6.89−6.86 (m, 1H), 6.76 (d, J = 7.0 Hz, 2H), 6.61 (d, J = 8.0 Hz, 1H), 5.49 (s, 1H), 5.20−5.18 (m, 1H), 5.01−4.98 (m, 1H), 4.02−3.98 (m, 1H), 3.41 (s, 3H), 2.97−2.94 (m, 1H), 2.44 (s, 3H). ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 171.0, 164.1, 144.1, 142.3, 141.8, 140.9, 136.9, 136.8, 135.8, 133.6, 129.3, 128.3, 128.2, 128.0, 127.8, 127.7, 127.5, 126.7, 126.4, 124.0, 123.4, 66.1, 62.5, 60.3, 42.9, 30.1, 21.5. IR (neat): ν_{max}3054, 2986, 2928, 1713, 1626, 1564, 1492, 1469, 1422, 1378, 1265, 1147, 1086, 1014, 933, 896, 746 cm^{-1}. HRMS (ESI-TOF): m/z [M + H]$^+$ calcd for $C_{34}H_{31}N_2O_4S$ 563.1999, found 563.2001.

*Benzyl (Z)-1′-Allyl-2-phenyl-2′-(tosylimino)spiro[cyclopentane-1,3′-indolin]-3-ene-3-carboxylate (**6ca**).* Following the general procedure, the reaction of **1c** (65.3 mg, 0.20 mmol) with **5a** (77.4 mg, 0.24 mmol) and tri(*p*-tolyl)phosphine (12.2 mg, 0.04 mmol) afforded **6ca** as a white solid. Yield: 93.0 mg (79%), mp 96−98 °C. ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.95 (d, J = 7.0 Hz, 2H), 7.28−7.16 (m, 7H), 7.01−6.88 (m, 7H), 6.74 (d, J = 6.0 Hz, 2H), 6.57 (d, J = 7.5 Hz, 1H), 5.74−5.69 (m, 2H), 5.20−5.17 (m, 2H), 5.10−5.07 (m, 1H), 5.00−4.98 (m, 1H), 4.62−4.60 (m, 1H), 4.29−4.18 (m, 2H), 2.97−2.93 (m, 1H), 2.43 (s, 3H). ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 170.5, 164.2, 144.2, 142.4, 140.8, 140.6, 136.9, 136.7, 135.7, 134.0, 130.2, 129.3, 128.3, 128.2, 128.1, 127.8, 127.7, 127.4, 126.7, 126.4, 124.0, 123.2, 118.6, 109.3, 66.1, 62.4, 59.9, 45.2, 43.1, 21.5. IR (neat):

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ν_{max}3054, 2986, 1712, 1561, 1422, 1265, 1148, 1087, 896, 738, 706, 666 cm^{-1}. HRMS (ESI-TOF): m/z $[M + H]^+$ calcd for $C_{36}H_{33}N_2O_4S$ 589.2156, found 589.2151.

*Benzyl (Z)-1′-Benzyl-2-phenyl-2′-(tosylimino)spiro-[cyclopentane-1,3′-indolin]-3-ene-3-carboxylate (**6da**).* Following the general procedure, the reaction of **1d** (75.3 mg, 0.20 mmol) with **5a** (77.4 mg, 0.24 mmol) and tri(*p*-tolyl)phosphine (12.2 mg, 0.04 mmol) afforded **6da** as a white solid. Yield: 96.0 mg (75%), mp 75−77 °C. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.92−7.90 (m, 2H), 7.32−7.19 (m, 10H), 7.13−7.11 (m, 2H), 7.01−6.81 (m, 7H), 6.65−6.63 (m, 2H), 6.58 (d, J = 8.0 Hz, 1H), 5.79−5.77 (m, 1H), 5.21−5.12 (m, 2H), 5.01−4.98 (m, 1H), 4.79−4.76 (m, 1H), 4.34−4.29 (m, 1H), 3.02−2.96 (m, 1H), 2.44 (s, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 171.0, 164.2, 144.1, 142.4, 140.7, 140.6, 137.1, 136.7, 135.8, 134.8, 134.3, 129.2, 128.6, 128.3$_2$, 128.2$_6$, 128.1, 127.8$_3$, 127.8$_1$, 127.7, 127.4, 127.0, 126.6, 126.5, 124.2, 123.2, 109.2, 66.1, 62.2, 59.6, 46.2, 43.4, 21.5. IR (neat): ν_{max}3054, 2986, 1715, 1637, 1422, 1265, 1147, 1083, 896, 740, 705, 665 cm^{-1}. HRMS (ESI-TOF): m/z $[M + H]^+$ calcd for $C_{40}H_{35}N_2O_4S$ 639.2312, found 639.2309.

*Benzyl (Z)-1′-Methyl-2-(thiophene-2-yl)-2′-(tosylimino)spiro-[cyclopentane-1,3′-indolin]-3-ene-3-carboxylate (**6bf**).* Following the general procedure, the reaction of **1b** (60.0 mg, 0.04 mmol) with **5f** (79.0 mg, 0.24 mmol) and tri(*p*-tolyl)phosphine (12.2 mg, 0.04 mmol) afforded **6bf** as a brown solid. Yield: 83.0 mg (73%), mp 66−68 °C. ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.92 (d, J = 8.0 Hz, 2H), 7.28−7.24 (m, 5H), 7.19−7.16 (m, 2H), 7.10−7.09 (m, 2H), 6.98 (d, J = 7.5 Hz, 1H), 6.94−6.90 (m, 2H), 6.75 (d, J = 7.5 Hz, 1H), 6.68−6.67 (m, 1H), 6.49 (d, J = 3.0 Hz, 1H), 5.60 (s, 1H), 5.21−5.19 (m, 1H), 5.05−5.02 (m, 1H), 3.84−3.80 (m, 1H), 3.50 (s, 3H), 3.00−2.96 (m, 1H), 2.42 (s, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 170.6, 163.7, 143.6, 142.4, 142.3, 140.9, 140.4, 137.2, 135.8, 133.0, 129.2, 128.5, 128.3, 127.9, 127.7, 127.3, 126.2, 126.0, 124.1, 124.0, 123.6, 108.9, 66.2, 62.2, 55.1, 43.2, 30.6, 21.5. IR (neat): ν_{max}3054, 2986, 1724, 1564, 1422, 1265, 1150, 1084, 896, 740, 705, 665 cm^{-1}. HRMS (ESI-TOF): m/z $[M + H]^+$ calcd for $C_{32}H_{29}N_2O_4S_2$ 569.1563, found 569.1563.

*Benzyl (Z)-1′-ethyl-5′-methoxy-2-phenyl-2′-(tosylimino)spiro-[cyclopentane-1,3′-indolin]-3-ene-3-carboxylate (**6ga**).* Following the general procedure, the reaction of **1g** (69.0 mg, 0.20 mmol) with **5a** (77.4 mg, 0.24 mmol) and tri(*p*-tolyl)phosphine (12.2 mg, 0.04 mmol) afforded **6ga** as a brown solid, Yield: 86.1 mg (71%), mp 96−98 °C. ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.95 (d, J = 8.5 Hz, 2H), 7.32−7.20 (m, 6H), 6.99−6.94 (m, 5H), 6.75 (d, J = 7.0 Hz, 2H), 6.67 (d, J = 2.0 Hz, 1H), 6.58−6.56 (m, 1H), 6.50 (d, J = 8.5 Hz, 1H), 5.58 (d, J = 1.5 Hz, 1H), 5.20−5.18 (m, 1H), 5.00−4.98 (m, 1H), 4.10−4.04 (m, 2H), 3.75−3.71 (m, 1H), 3.68 (s, 3H), 2.94−2.91 (m, 1H), 2.42 (s, 3H), 1.22 (t, J = 7.0 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 169.7, 164.1, 156.4, 144.2, 142.1, 141.2, 136.9, 136.8, 135.7, 135.5, 134.2, 129.2, 128.1, 127.9, 127.7, 127.5, 126.7, 126.3, 113.0, 111.3, 109.1, 66.1, 62.4, 60.0, 56.0, 43.1, 37.9, 21.5, 12.1. IR (neat): ν_{max}3054, 2986, 1712, 1559, 1493, 1422, 1265, 1148, 1087, 1028, 896, 736, 706 cm^{-1}. HRMS (ESI-TOF): m/z $[M + H]^+$ calcd for $C_{36}H_{35}N_2O_5S$ 607.2261, found 607.2262.

*3-Benzyl 1′-(tert-Butyl) (Z)-2-(4-Methoxyphenyl)-2′-(tosylimino)-spiro[cyclopentane-1,3′-indolin]-3-ene-1′,3-dicarboxylate (**6hb**).* Following the general procedure, the reaction of **1h** (77.2 mg, 0.04 mmol) with **5b** (84.5 mg, 0.24 mmol) and tri(*p*-tolyl)phosphine (12.2 mg, 0.04 mmol) afforded **6hb** as a white solid. Yield: 114.0 mg (84%), mp 63−65 °C. ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.88 (d, J = 8.0 Hz, 2H), 7.40 (d, J = 8.0 Hz, 1H), 7.26−7.24 (m, 5H), 7.10−7.07 (m, 2H), 6.99 (d, J = 6.0 Hz, 2H), 6.87−6.84 (m, 1H), 6.77−6.73 (m, 3H), 6.56 (d, J = 8.5 Hz, 2H), 5.17 (d, J = 12.5 Hz, 1H), 4.95−4.90 (m, 2H), 3.70 (s, 3H), 3.29−3.26 (m, 1H), 2.88−2.84 (m, 1H), 2.40 (s, 3H), 1.66 (s, 9H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 169.1, 163.8, 158.6, 148.9, 143.1, 142.8, 140.0, 139.9, 136.5, 135.8, 130.8, 129.3, 129.2, 129.1, 128.5, 128.3, 127.8, 127.6, 126.7, 124.1$_7$, 124.1$_5$, 113.2$_0$, 113.1$_6$, 86.7, 66.0, 62.3, 61.3, 55.1, 43.4, 27.7, 21.5. IR (neat): ν_{max}3034, 2933, 2834, 1754, 1715, 1631, 1596 1511, 1461, 1318, 1282, 1176, 1084, 1031, 810, 697 cm^{-1}. HRMS (ESI-TOF): m/z $[M + H]^+$ calcd for $C_{39}H_{39}N_2O_7S$ 679.2472, found 679.2473.

*vii. Synthesis tert-Butyl 3-((2Z,4E)-1-Ethoxy-1-oxo-5-phenylpenta-2,4-dien-2-yl)-2-((4-methylphenyl)sulfonamido)-1H-indole-1-carboxylate (**7ha**).* A Schlenk tube was charged with 2-sulfonamido indole **1h** (77.2 mg, 0.20 mmol), δ-acetoxy allenoate **2a** (62.4 mg, 0.24 mmol), and toluene (2.0 mL). Subsequently, Ph_3P (10.5 mg, 0.04 mmol) was added to the reaction mixture at rt (25 °C), the mixture was stirred at the same temperature for 36 h, and the progress of the reaction was monitored using TLC. After completion of the reaction, the mixture was quenched by addition of water (10 mL). The aqueous layer was extracted with ethyl acetate (3 × 10 mL). Then the combined organic layer was washed with brine solution (20.0 mL), dried over anhydrous Na_2SO_4, and concentrated under reduced pressure. The crude product was then purified by silica gel column chromatography using ethyl acetate/hexane (2:8) as the eluent to obtain **7ha** as a white solid. Yield: 91.4 mg (78%), mp 103−105 °C. ^{1}H NMR (500 MHz, $CDCl_3$): δ 8.19 (s, 1H), 7.89 (d, J = 8.0 Hz, 1H), 7.51 (d, J = 8.0 Hz, 2H) 7.32−7.24 (m, 8H), 7.11−7.06 (m, 3H), 6.95 (d, J = 8.0 Hz, 1H), 6.69 (d, J = 11.5 Hz, 1H), 5.99 (d, J = 15.0 Hz, 1H), 4.18 (q, J = 7.0 Hz, 2H), 2.32 (s, 3H), 1.64 (s, 9H), 1.27 (t, J = 7.0 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 167.2, 150.5, 143.7, 142.3, 140.1, 139.3, 136.6, 133.4, 129.5, 129.2, 128.7, 128.2, 127.8$_4$, 127.7$_6$, 127.6, 127.3, 125.2, 123.5, 122.4, 120.8, 115.3, 115.2, 86.0, 60.2, 28.1, 21.5, 14.3. IR (neat): ν_{max} 3235, 2971, 1701, 1619, 1452, 1362, 1274, 1186, 1037, 700 cm^{-1}. HRMS (ESI-TOF): m/z $[M + H]^+$ calcd for $C_{33}H_{35}N_2O_6S$ 587.2210, found: 587.2220.

*viii. Scale-Up Experiments for **3aa**, **4aa**, and **6aa**. Synthesis of **3aa**.* A 25 mL RB flask was charged with **1a** (314.0 mg, 1.0 mmol), **2a** (312.0 mg, 1.20 mmol), and toluene (10.0 mL). Subsequently, Ph_3P (52.5 mg, 0.20 mmol) was added to the reaction mixture at rt, and the mixture was stirred at the same temperature for 36 h. Then it was quenched by addition of water (10 mL). The aqueous layer was extracted with ethyl acetate (3 × 10 mL). Then the combined organic layer was washed with brine (20 mL), dried over anhydrous Na_2SO_4, and concentrated under reduced pressure. The crude product was purified by silica gel column chromatography using ethyl acetate/hexane (1:9) as the eluent to obtain **3aa** (406.2 mg; 79%).

*Synthesis of **4aa**.* A 25 mL RB flask was charged with **1a** (314.0 mg, 1.0 mmol), Ph_3P (52.5 mg, 0.20 mmol), cesium carbonate (651.6 mg, 2.0 mmol), and toluene (5.0 mL). Subsequently, **2a** (312.0 mg, 1.20 mmol) in toluene (5.0 mL) was added gradually over a period of 45 min at 80 °C (oil bath), and the mixture was stirred at the same temperature for 36 h. Then it was quenched by addition of water (10 mL). The aqueous layer was extracted with ethyl acetate (3 × 10 mL). Then the combined organic layer was washed with brine (20 mL), dried over anhydrous Na_2SO_4, and concentrated under reduced pressure. The crude product was purified by silica gel column chromatography using ethyl acetate/hexane (1:9) as the eluent to obtain **4aa** (369.0 mg; 72%).

*Synthesis of **6aa**.* A 25 mL RB flask was charged with **1a** (314.0 mg, 1.0 mmol), tri(*p*-tolyl)phosphine (61.0 mg, 0.20 mmol), and toluene (5.0 mL). Subsequently, **5a** (386.5 mg, 1.20 mmol) in toluene (5.0 mL) was added gradually over a period of 45 min at 80 °C (oil bath), and the mixture was stirred at the same temperature for 36 h. Then it was quenched by addition of water (10 mL). The aqueous layer was extracted with ethyl acetate (3 × 10 mL). Then the combined organic layer was washed with brine (20 mL), dried over anhydrous Na_2SO_4, and concentrated under reduced pressure. The crude product was purified by silica gel column chromatography using ethyl acetate/hexane (1:4) as the eluent to obtain **6aa** (432.0 mg; 75%).

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.joc.1c01137.

Experimental details, characterization data, X-ray crystallographic data, ORTEPs of compounds **3fa**, **4aa**, **4ad**, **6aa**, and **6ae** (Figures S1−S5), and ^{1}H/ $^{13}C\{^1H\}$ NMR spectra (PDF)

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CCDC 2064415−2064419 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

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